WSQ · DATA VISUALISATION

Data Analytics and Visualization with Power BI

Turn data into trusted decisions

Course Code: TGS-2020505444
Version: v15.0 · 20 August 2026
Trainer: _____

Tertiary Infotech Academy Pte Ltd
UEN: 201200696W



Digital Attendance (Mandatory)

1

AM

Complete SSG digital attendance at the start

2

PM

Complete the afternoon attendance

3

Assessment

Complete the assessment attendance before the papers

4

Evidence

Use the trainer-provided TRAQOM QR code

About the Trainer · Approved Facilitator Profile

TF

Approved Facilitator

Profile template · complete before class

NAME	[Trainer name]
TITLE / DESIGNATION	[Current role]
QUALIFICATIONS	[Relevant credentials]
AREAS OF EXPERTISE	[Power BI and data specialisms]
TRAINING & EXPERIENCE	[Industry and facilitation experience]
CONTACT	[Professional contact]

About the Trainer · Dr Alfred Ang



Dr Alfred Ang

Founder · Trainer · Technologist

NAME	Dr Alfred Ang
TITLE / DESIGNATION	Founder and Lead Trainer
QUALIFICATIONS	Adult educator and technology practitioner
AREAS OF EXPERTISE	AI, analytics, automation and digital transformation
TRAINING & EXPERIENCE	Industry-led, applied learning for working professionals
CONTACT	enquiry@tertiaryinfotech.com

Let’s Know Each Other

01 Name and role

02 What decision do you support with data?

03 Your Power BI experience

04 One outcome you want from the course

Ground Rules

1

Participate

Ask, test and explain your reasoning

2

Respect

One conversation; challenge ideas, not people

3

Focus

Phones silent and confidential data protected

4

Time

Return promptly; 75% attendance is required

WSQ ALIGNMENT

Skills Framework

1

Data Visualisation

TSC MED-DIT-4006-1.1



SIGNAL

2

Knowledge

K1–K5: tools, methods, dashboards, displays and mapping

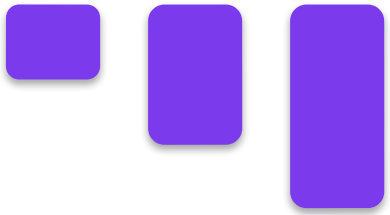


SIGNAL

3

Abilities

A1–A6: select, design, build, review and align findings



SIGNAL

Knowledge and Ability Map

1

K1–K2

Tools, techniques and visual methods

2

K3–K5

Dashboards, display features and strategic mapping

3

A1–A3

Select fit-for-purpose tools, present trends and build dashboards

4

A4–A6

Review dynamic displays, design UX and align findings with SMEs

WHAT YOU WILL ACHIEVE

Learning Outcomes

1

LO1

Perform data transformation techniques using Power BI

2

LO2

Create data visualizations that portray patterns, trends and correlations

3

LO3

Develop dashboards and scorecards for decision monitoring

4

LO4

Create dynamic and interactive data visualizations

Learning Outcomes · Model and Analyze

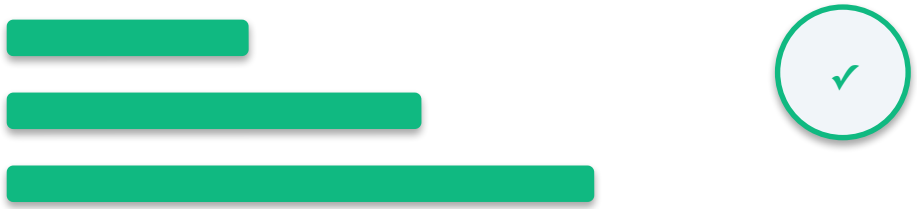
LO5

Create a data model for strategic visualization and mapping



LO6

Perform data analytics and queries with DAX



TWO-DAY JOURNEY

Course Outline

1

Day 1

Power Query · visuals · reports · dashboards



SIGNAL

2

Day 2

Interactions · modeling · DAX · capstone

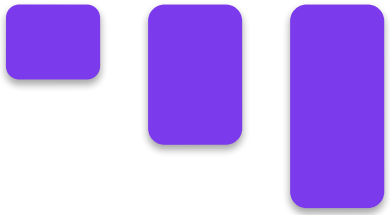


SIGNAL

3

Assessment

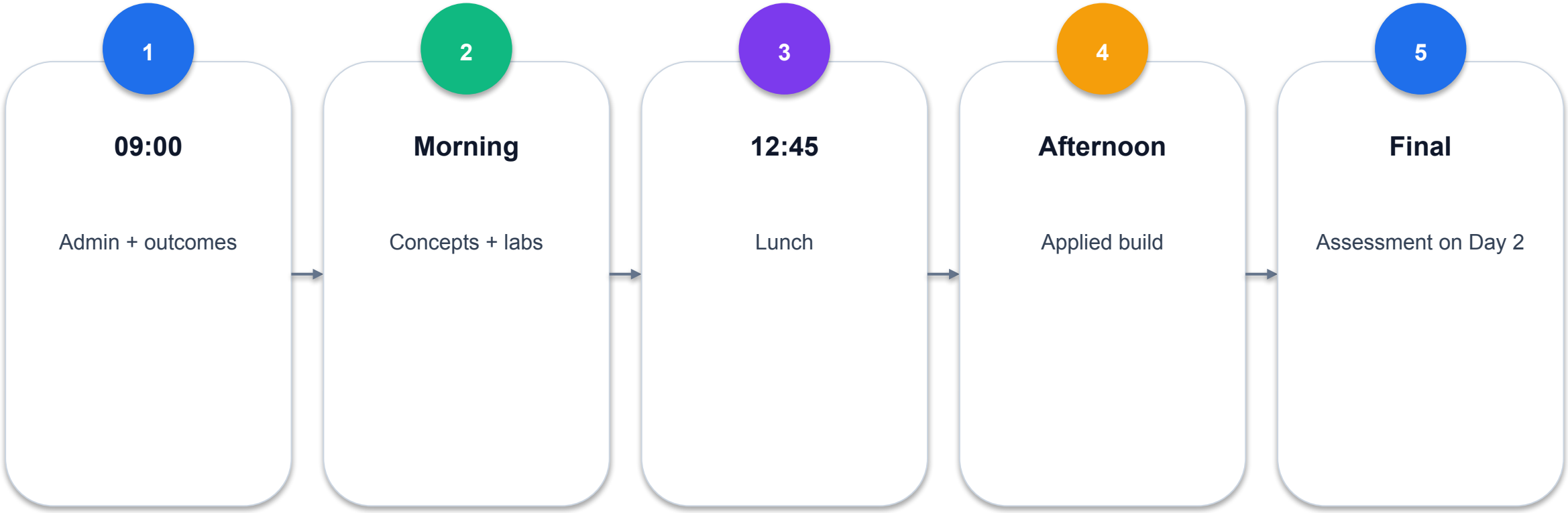
Written Assessment 60 min · Practical Performance 90 min



SIGNAL

9:00 AM – 6:00 PM

Lesson Plan



Briefing for Assessment

01 Individual work; no discussion

02 Open book: slides, Learner Guide and approved materials only

03 Phones and other materials under the table

04 Use your own words and retain required evidence

05 Submit on the LMS and follow the appeal/reassessment process

Assessment and Funding

1

Written Assessment

5 open-ended questions · K1–K5 · 60 min



SIGNAL

2

Practical Performance

5 applied tasks · A1–A6 · 90 min

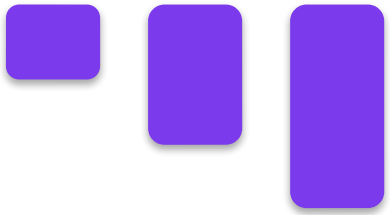


SIGNAL

3

Funding criteria

At least 75% attendance and Competent outcome



SIGNAL

Assessment Flow

1

TRAQOM

Scan the QR code on the LMS

2

Attendance

Complete assessment digital attendance

3

Assessment

Sit WA, then PP; open book

4

Submit

Upload answers on the LMS

5

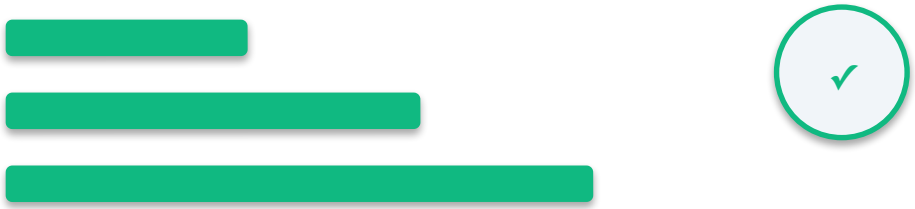
Sign

Complete the Assessment Summary Record

Access Course Materials

Option A · LMS

Sign in to https://lms-tms.tertiaryinfotech.com/
Open the course record → Courseware → download the current files.



Option B · GitHub

Download ZIP or clone https://github.com/tertiarycourses/TGS-2020505444-Data-Analytics-and-Visualization-with-Power-BI.git

Each lab folder is self-contained.



https://lms-tms.tertiaryinfotech.com/ · https://github.com/tertiarycourses/TGS-2020505444-Data-Analytics-and-Visualization-with-Power-BI

TOPIC

01

Get Started with Power BI and Power Query

K1 · A1 · LO1

Concepts on slides · detailed procedures in the Learner Guide and lab folders

CONCEPT

From data to action

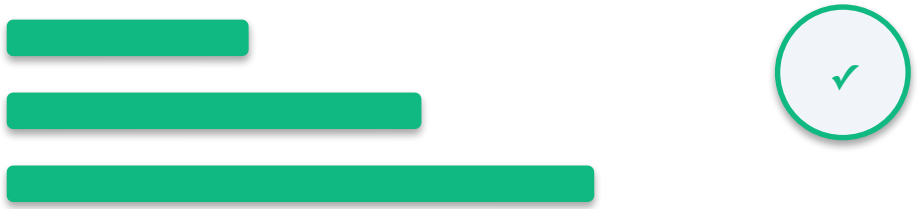
Core idea

Descriptive, diagnostic, predictive, prescriptive and cognitive analytics support different decisions.



Decision lens

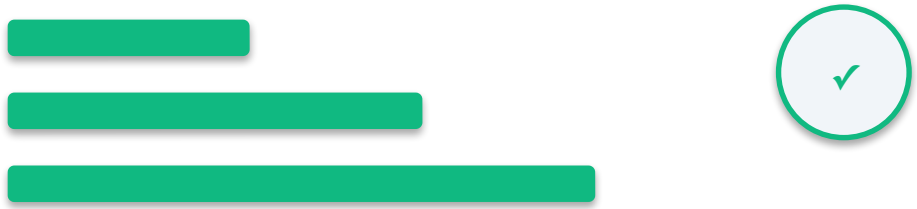
Use the idea to support a real stakeholder decision.



Roles in the data team · The analyst workflow

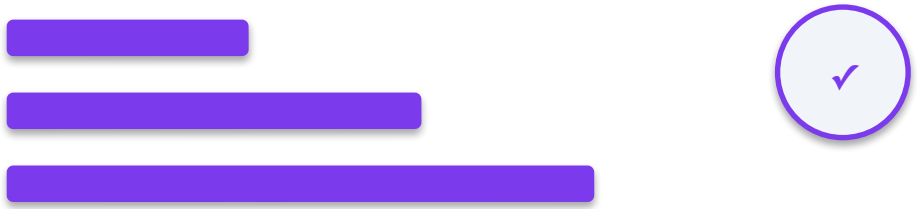
Roles in the data team

Business experts frame the question; analysts transform evidence; engineers govern dependable data flows.



The analyst workflow

Prepare → model → visualize → analyze → publish → monitor; each stage leaves evidence for review.



CONCEPT

Power BI ecosystem

1

What

Desktop authors reports and semantic models.

2

Why

Service distributes, refreshes, governs and collaborates.

3

Evidence

Use a defined metric, source and filter state.

4

Action

Translate the finding into an owner and next step.

CONCEPT

Dataset, semantic model, report, dashboard

1

Meaning

Separate reusable data logic from multi-page analysis and single-page monitoring.



SIGNAL

2

Decision use

Use the idea to support a real stakeholder decision.

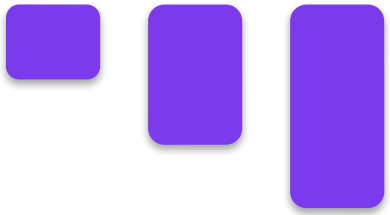


SIGNAL

3

Quality check

Confirm grain, filter context and whether the display could be misread.



SIGNAL

WHY IT MATTERS

Desktop views

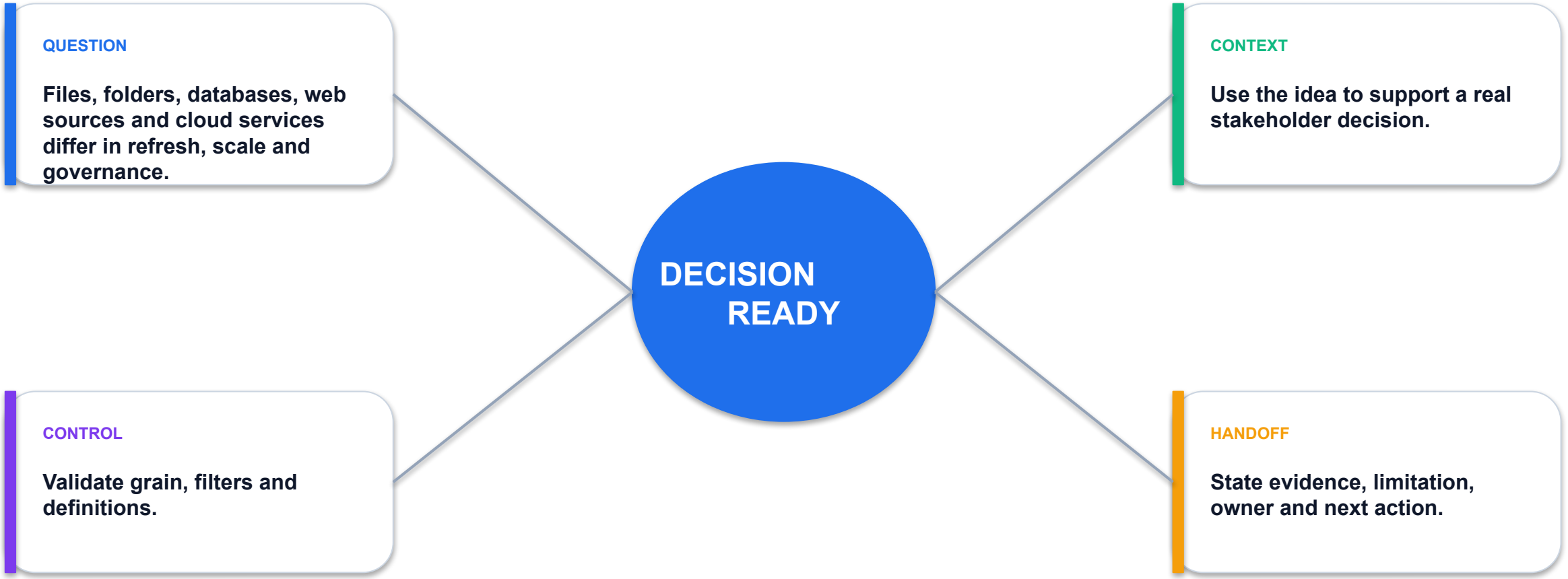
> **Report view composes visuals, Data view inspects loaded tables, and Model view manages relationships.**

Use the idea to support a real stakeholder decision.

QUESTION	What must be decided?
EVIDENCE	What supports the claim?
ACTION	Who acts next?

CONCEPT MAP

Connection choices



CONCEPT

Import or DirectQuery

1

What

Import favors speed and modeling freedom.

2

Why

DirectQuery favors source currency but adds dependency and latency.

3

Evidence

Use a defined metric, source and filter state.

4

Action

Translate the finding into an owner and next step.

Navigator decisions

1	**Frame**	Select only required tables and preview structure before loading to reduce noise and model size.
2	**Inspect**	Use the idea to support a real stakeholder decision.
3	**Test**	Check the result under a second filter or scenario.
4	**Communicate**	State finding, evidence, limitation and action.

CONCEPT

Power Query Editor

1

Meaning

Applied Steps make transformations visible, ordered, repeatable and auditable.



SIGNAL

2

Decision use

Use the idea to support a real stakeholder decision.

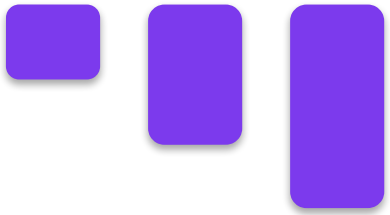


SIGNAL

3

Quality check

Confirm grain, filter context and whether the display could be misread.



SIGNAL

Promote headers and remove rows

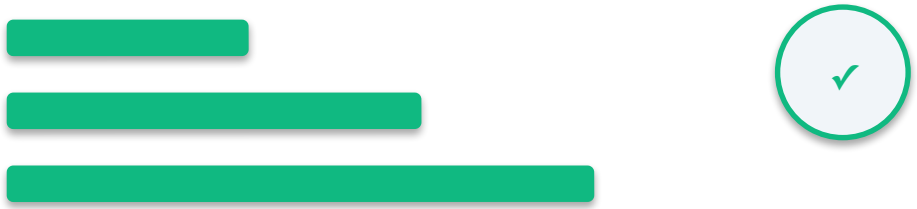
Core idea

Fix structural metadata before analytical logic so column names and data boundaries are reliable.



Decision lens

Use the idea to support a real stakeholder decision.



CONCEPT

Choose and rename columns

1

What

Keep only decision-relevant fields and give them business-friendly names.

2

Why

Use the idea to support a real stakeholder decision.

3

Evidence

Use a defined metric, source and filter state.

4

Action

Translate the finding into an owner and next step.

CONCEPT

Pivot and unpivot

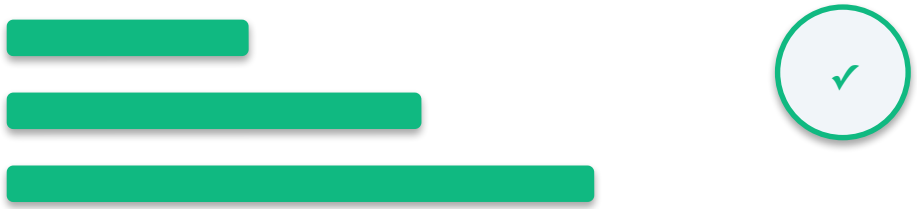
Core idea

Wide layouts are human-friendly.



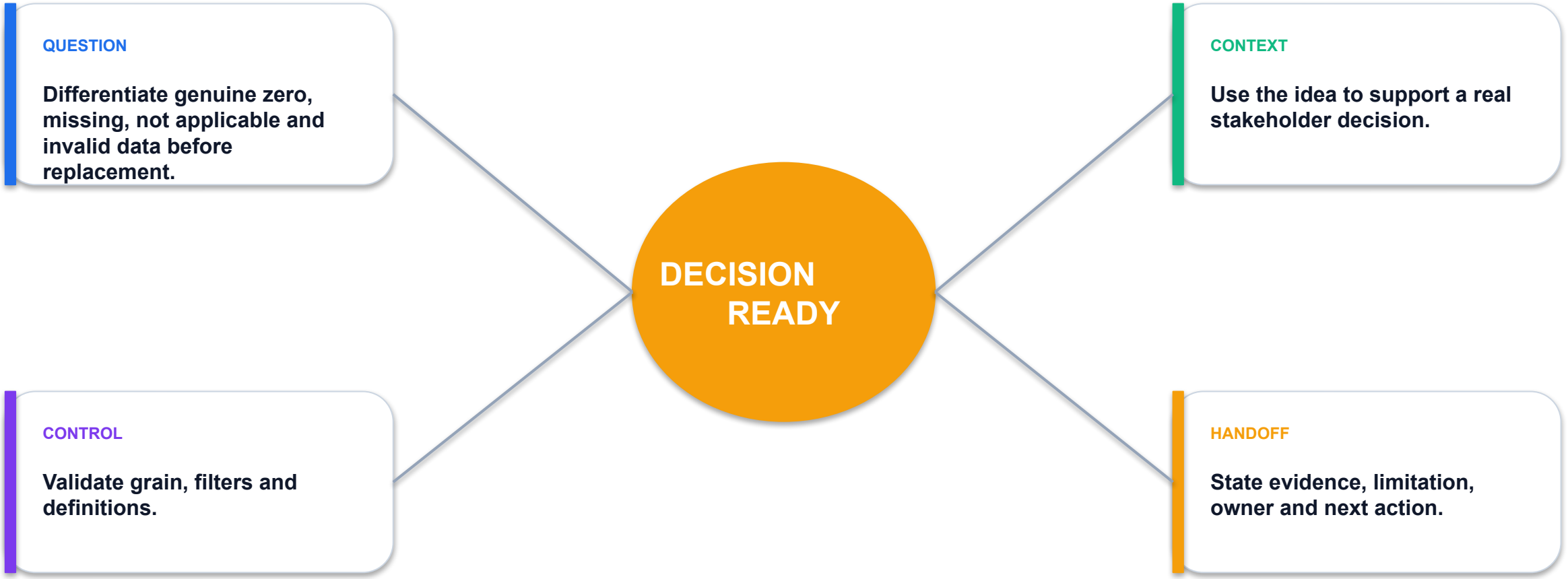
Decision lens

long tidy layouts are usually better for filtering, aggregation and visuals.



CONCEPT MAP

Replace values, nulls and errors



CONCEPT

Remove duplicates

1

Meaning

Define the business key before deduplication.



SIGNAL

2

Decision use

repeated rows are not always duplicates.

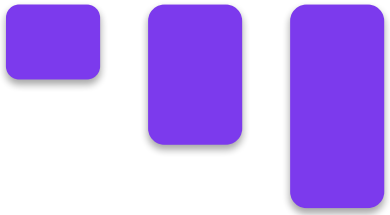


SIGNAL

3

Quality check

Confirm grain, filter context and whether the display could be misread.



SIGNAL

CONCEPT

Data types and locale

1

What

Types determine aggregation and date behavior.

2

Why

locale determines how text values are interpreted.

3

Evidence

Use a defined metric, source and filter state.

4

Action

Translate the finding into an owner and next step.

CONCEPT

Append queries

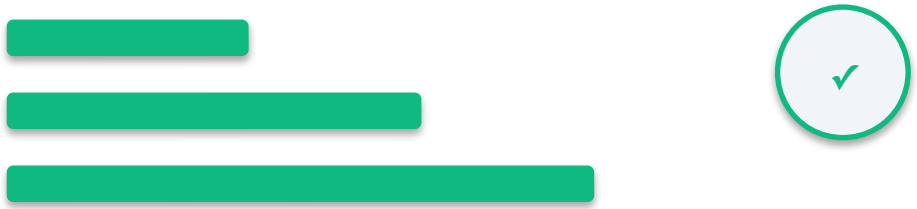
Core idea

Stack tables with the same grain and schema, then reconcile before/after row counts.



Decision lens

Use the idea to support a real stakeholder decision.



Merge queries

1. **Frame** — Join tables by validated keys.
2. **Inspect** — left outer, inner and full outer joins answer different integrity questions.
3. **Test** — Check the result under a second filter or scenario.
4. **Communicate** — State finding, evidence, limitation and action.

CONCEPT

Column profiling

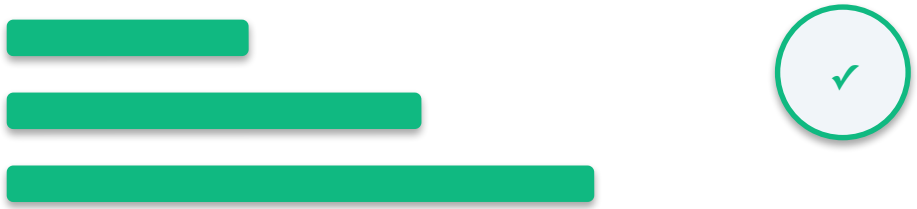
Core idea

Quality, distribution and profile views surface nulls, errors, outliers and unexpected cardinality.



Decision lens

Use the idea to support a real stakeholder decision.



CONCEPT

M and Advanced Editor

1

Meaning

Power Query records transformations as M.



SIGNAL

2

Decision use

readable steps improve reuse and troubleshooting.

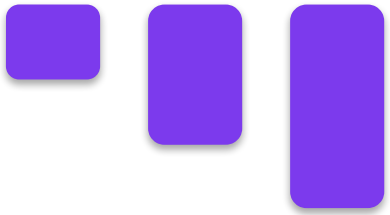


SIGNAL

3

Quality check

Confirm grain, filter context and whether the display could be misread.



SIGNAL

Lab 1 · Connect, Inspect and Profile an Excel Dataset

FOUNDATION

01

LO1 · K1 · A1

YOU'LL PRODUCE

A profiled query with correct headers, types and documented quality issues

FILES

Retail_Intake.xlsx · starter.pbix · README.md/PDF

PASS WHEN

All required tables load; column quality is reviewed; no unintended errors remain.

PROCEDURE Learner Guide + lab README

Lab 2 · Clean, Transform, Append and Merge

FOUNDATION

02

LO1 · K1 · A1

YOU'LL PRODUCE

A repeatable Power Query transformation pipeline with an audit trail

FILES

Retail_Quality_Issues.xlsx · starter.pbix · README.md/PDF

PASS WHEN

Headers, types, nulls and duplicates are corrected; append and merge row counts reconcile.

PROCEDURE Learner Guide + lab README

TOPIC

02

Create Data Visualizations and Reports

K2 · A2 · LO2

Concepts on slides · detailed procedures in the Learner Guide and lab folders

Start with the question

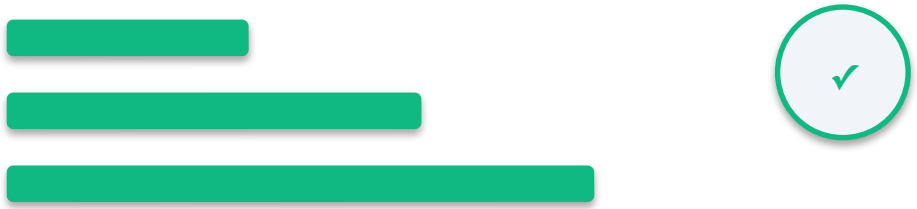
Core idea

Choose a visual only after naming the comparison, trend, distribution, relationship or composition to reveal.



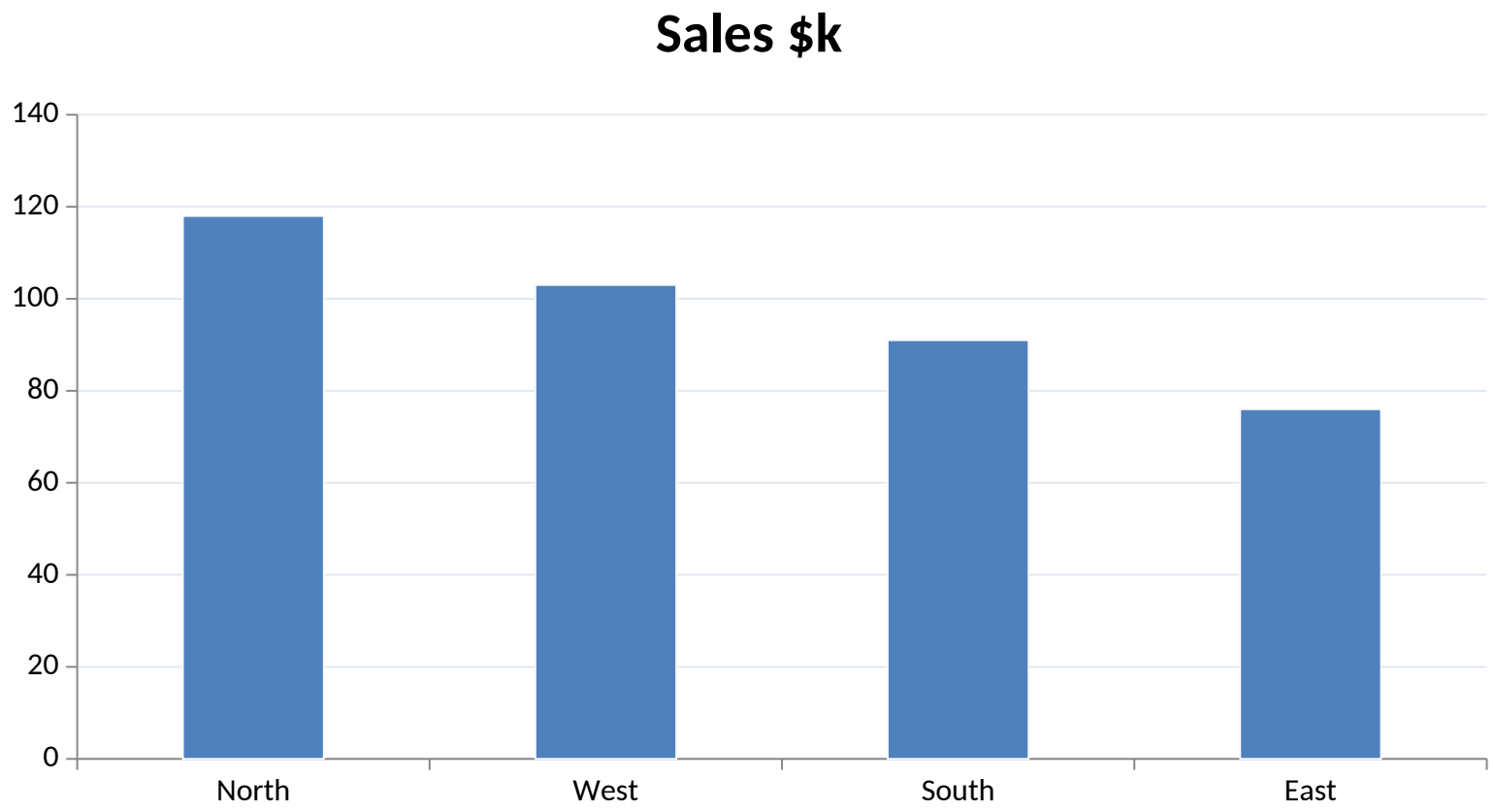
Decision lens

Use the idea to support a real stakeholder decision.



EDITABLE POWERPOINT CHART

Column and bar charts



WHAT THE DATA SHOWS

A common baseline and descending order make the ranking immediately comparable.

Chart is native and editable in PowerPoint.

Line and area charts · Cards, KPIs and gauges

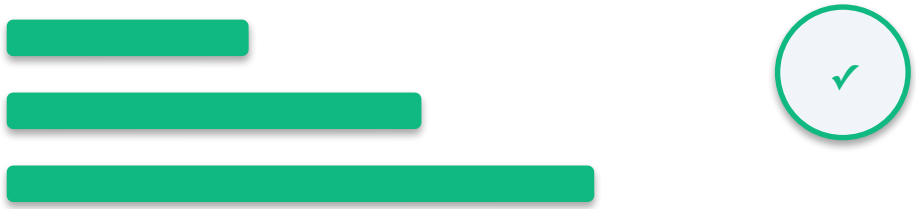
Line and area charts

Use lines for change over ordered time; use area sparingly when magnitude and composition both matter.

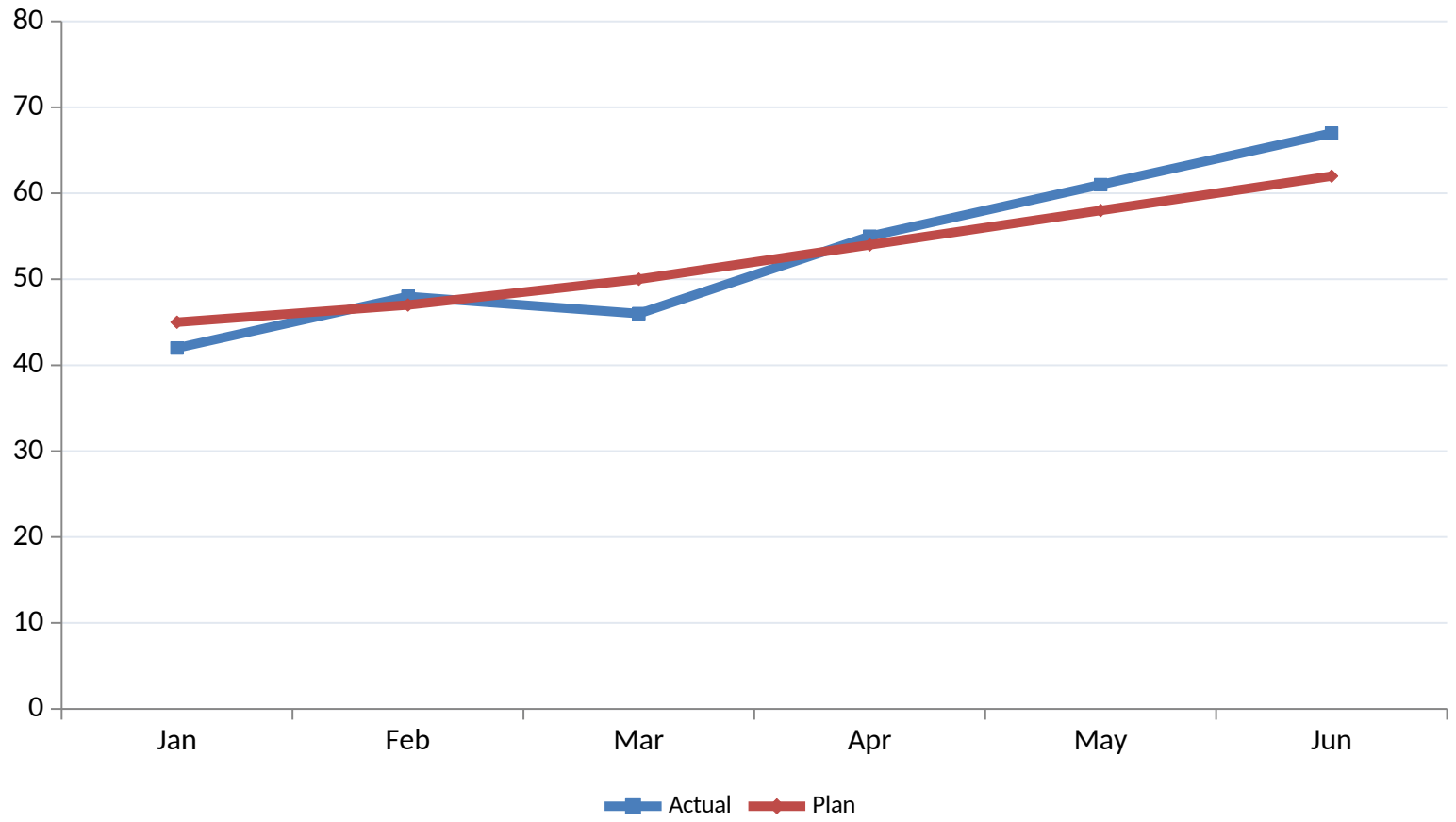


Cards, KPIs and gauges

Cards state a value, KPIs add direction and targets, and gauges show progress toward one bounded goal.



Combo charts



WHAT THE DATA SHOWS

The editable visual links a measured pattern to a specific review question.

Chart is native and editable in PowerPoint.

WHY IT MATTERS

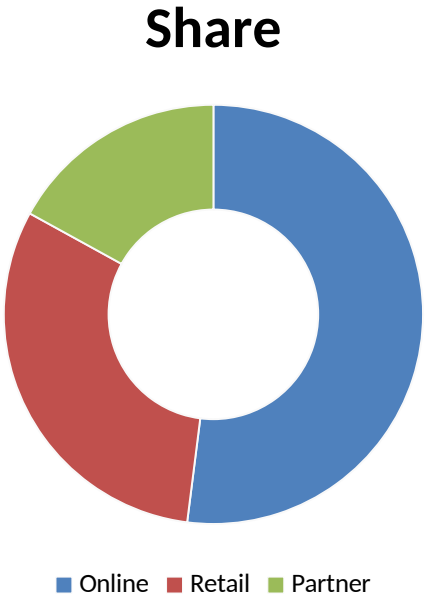
Decomposition tree

> **Let users drill across candidate dimensions to explain a result without predefining one path.**

Use the idea to support a real stakeholder decision.

QUESTION	What must be decided?
EVIDENCE	What supports the claim?
ACTION	Who acts next?

Pie, doughnut and funnel



PART TO WHOLE

Few categories; total = 100%



ORDERED STAGES

Show conversion, not categories

USE WHEN

The whole is meaningful

Categories are few

Stages have a fixed order

CONCEPT

Maps and choropleths

1

What

Use symbols for magnitude and filled regions for rates.

2

Why

normalize before comparing unequal geographies.

3

Evidence

Use a defined metric, source and filter state.

4

Action

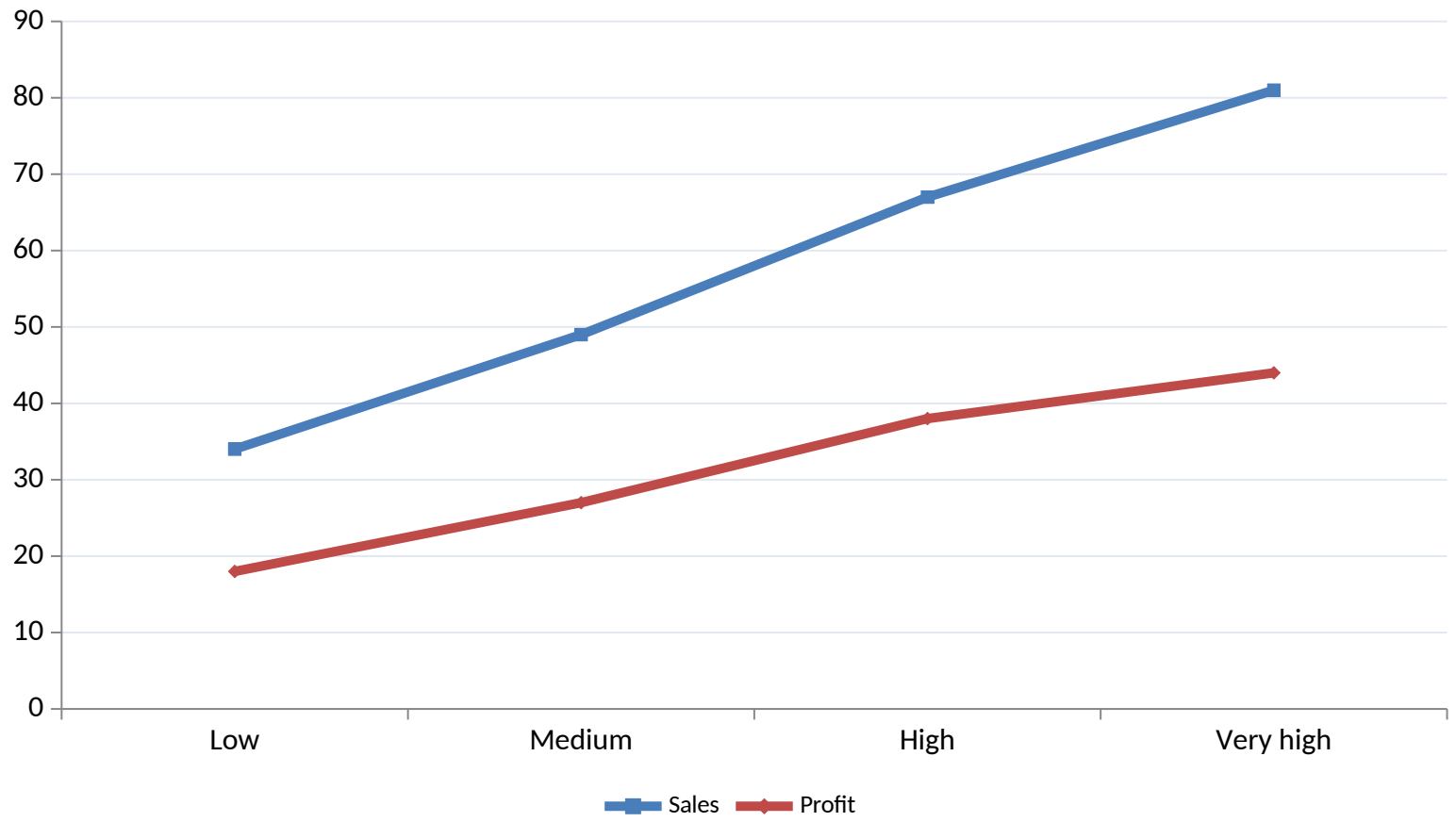
Translate the finding into an owner and next step.

Tables and matrices

1. **Frame** — Tables preserve precise records.
2. **Inspect** — matrices add hierarchy, subtotals and cross-tab analysis.
3. **Test** — Check the result under a second filter or scenario.
4. **Communicate** — State finding, evidence, limitation and action.

EDITABLE POWERPOINT CHART

Scatter and bubble charts



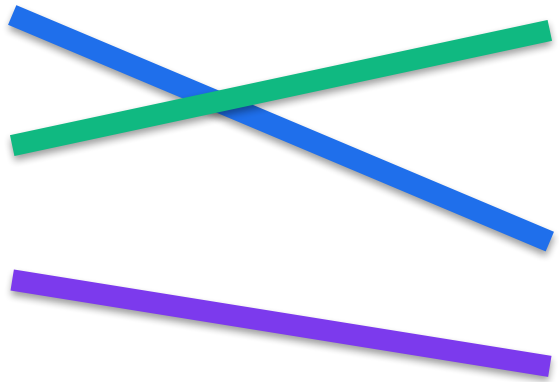
WHAT THE DATA SHOWS

Both measures rise together in this sample, but the display supports association—not a causal claim.

Chart is native and editable in PowerPoint.

Ribbon, treemap and waterfall

RIBBON



Rank changes

TREEMAP



Hierarchy + area

WATERFALL



Additive bridge

CONCEPT

Q&A and smart narratives

1

What

Natural-language and generated summaries accelerate exploration but still require human validation.

2

Why

Use the idea to support a real stakeholder decision.

3

Evidence

Use a defined metric, source and filter state.

4

Action

Translate the finding into an owner and next step.

Numeric, grid and performance visuals

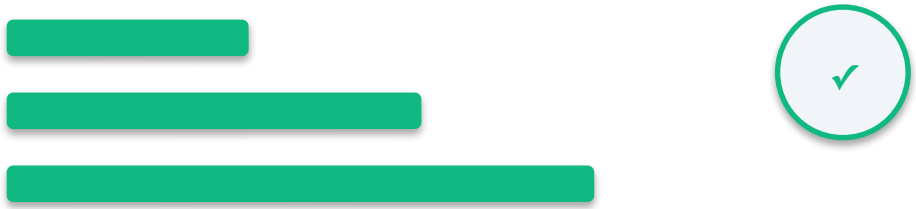
Core idea

Match information density to the decision: glanceable status, precise lookup or target tracking.



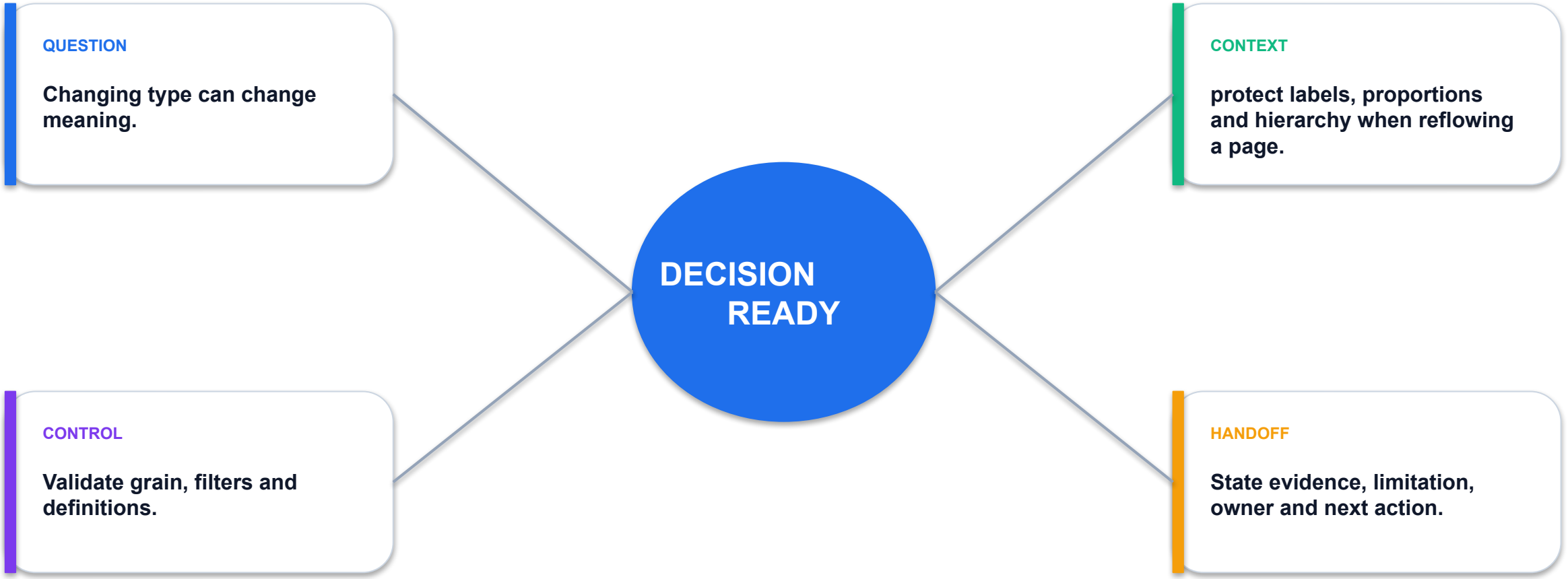
Decision lens

Use the idea to support a real stakeholder decision.



CONCEPT MAP

Change and resize visuals



CONCEPT

Format for comprehension

1

Meaning

Use consistent color roles, restrained emphasis and direct labels to reduce decoding effort.



SIGNAL

2

Decision use

Use the idea to support a real stakeholder decision.

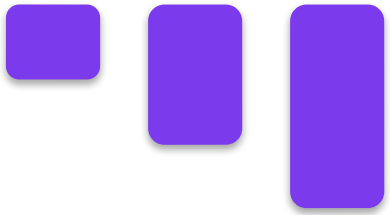


SIGNAL

3

Quality check

Confirm grain, filter context and whether the display could be misread.



SIGNAL

CONCEPT

Titles and backgrounds

1

What

Insight titles state the takeaway.

2

Why

backgrounds should organize, not decorate.

3

Evidence

Use a defined metric, source and filter state.

4

Action

Translate the finding into an owner and next step.

CONCEPT

Tooltips

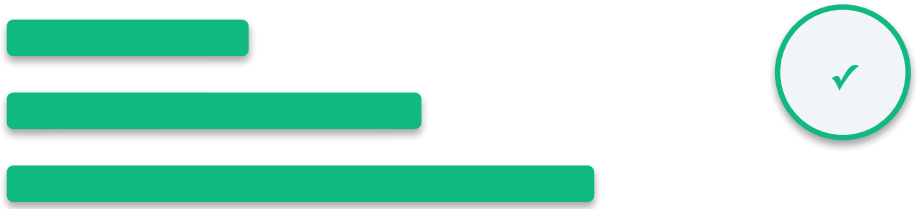
Core idea

Use tooltips for supporting context, definitions and secondary measures without crowding the main canvas.



Decision lens

Use the idea to support a real stakeholder decision.



DECISION STEPS

Drill down and expand

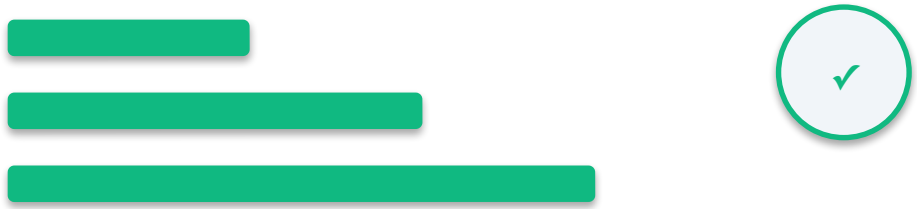
1	**Frame**	Hierarchies support movement between levels.
2	**Inspect**	keep the current level and path visible.
3	**Test**	Check the result under a second filter or scenario.
4	**Communicate**	State finding, evidence, limitation and action.

CONCEPT

Themes

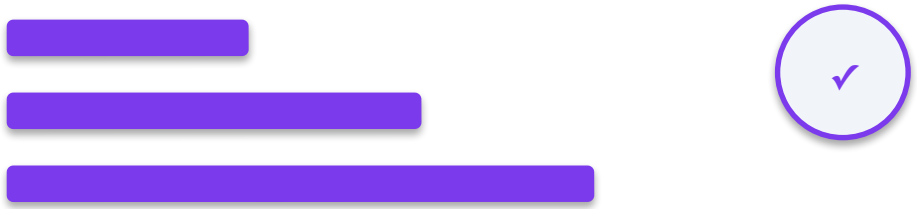
Core idea

A theme standardizes color, type and defaults so reports remain coherent and maintainable.



Decision lens

Use the idea to support a real stakeholder decision.



CONCEPT

Accessibility

1

Meaning

Reading order, alt text, contrast, keyboard focus and non-color cues make insight available to more users.



SIGNAL

2

Decision use

Use the idea to support a real stakeholder decision.

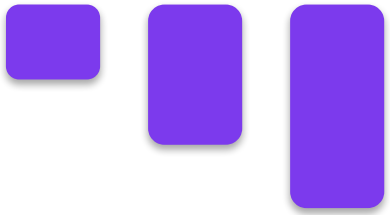


SIGNAL

3

Quality check

Confirm grain, filter context and whether the display could be misread.



SIGNAL

HANDS-ON LAB

Lab 3 · Choose Visuals for Trends and Correlations

INTERMEDIATE

03

LO2 · K2 · A2

YOU'LL PRODUCE

A one-page analytical report using fit-for-purpose native visuals

FILES

Sales_Trends.xlsx · starter.pbix · README.md/PDF

PASS WHEN

Each visual answers a stated question; encodings are accurate; titles express insight.

PROCEDURE Learner Guide + lab README

Lab 4 · Design an Accessible Report Story

INTERMEDIATE

04

LO2 · K2 · A2

YOU'LL PRODUCE

A polished report page with hierarchy, theme, tooltips and accessible navigation

FILES

Customer_Experience.xlsx · starter.pbix · README.md/PDF

PASS WHEN

The page has a clear reading order, consistent formatting, alt text and no misleading scale.

PROCEDURE Learner Guide + lab README

TOPIC

03

Develop Dashboards and Scorecards

K3 · A3 · LO3

Concepts on slides · detailed procedures in the Learner Guide and lab folders

CONCEPT

Report versus dashboard

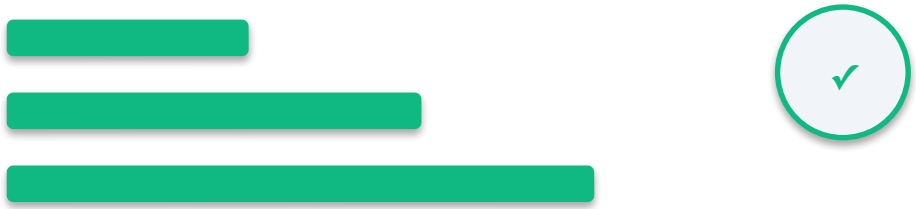
Core idea

Reports are multi-page interactive analysis.



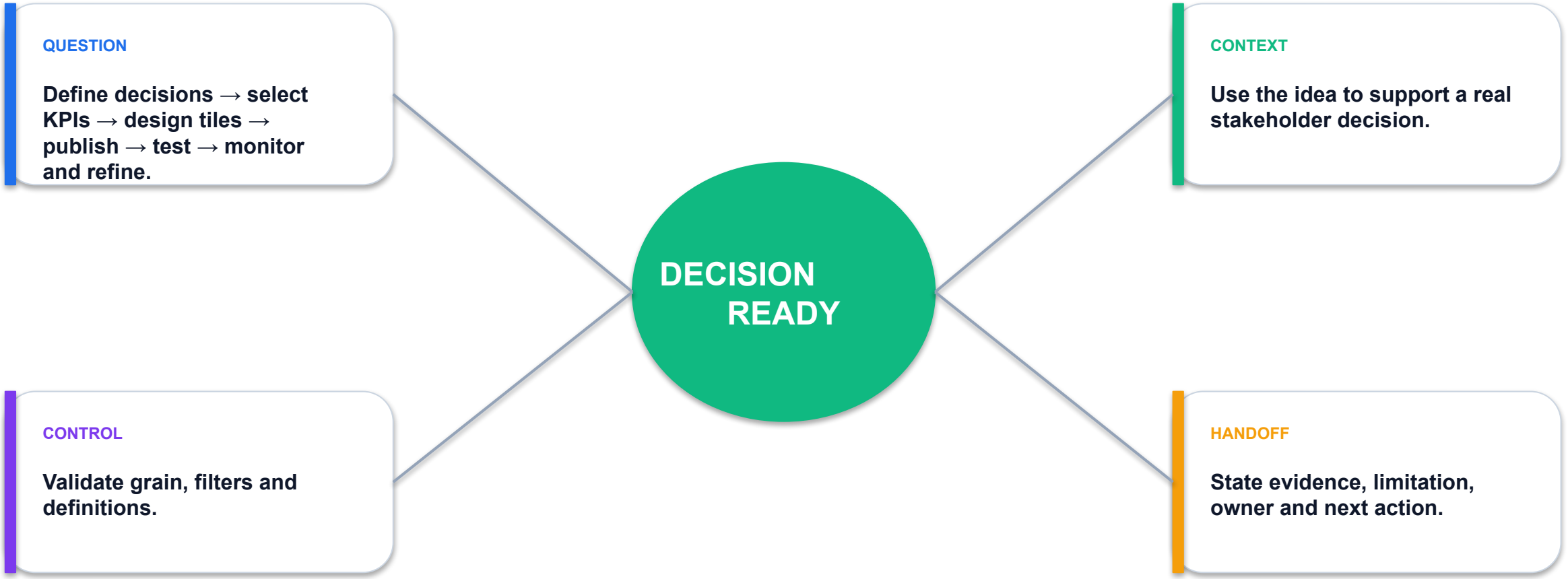
Decision lens

dashboards are one-page monitoring surfaces in Power BI Service.



CONCEPT MAP

Dashboard development cycle



WHY IT MATTERS

Pin a visual

> **Pinned tiles create a stable monitoring surface while preserving a path back to the source report.**

Use the idea to support a real stakeholder decision.

QUESTION	What must be decided?
EVIDENCE	What supports the claim?
ACTION	Who acts next?

CONCEPT

Pin a live page

1

What

A live page retains report interactions.

2

Why

use it when monitoring needs more than independent tiles.

3

Evidence

Use a defined metric, source and filter state.

4

Action

Translate the finding into an owner and next step.

CONCEPT

Combine multiple reports

1

Meaning

A dashboard can assemble governed signals from several reports into one executive view.



SIGNAL

2

Decision use

Use the idea to support a real stakeholder decision.

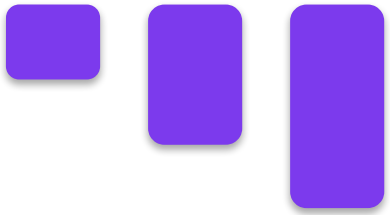


SIGNAL

3

Quality check

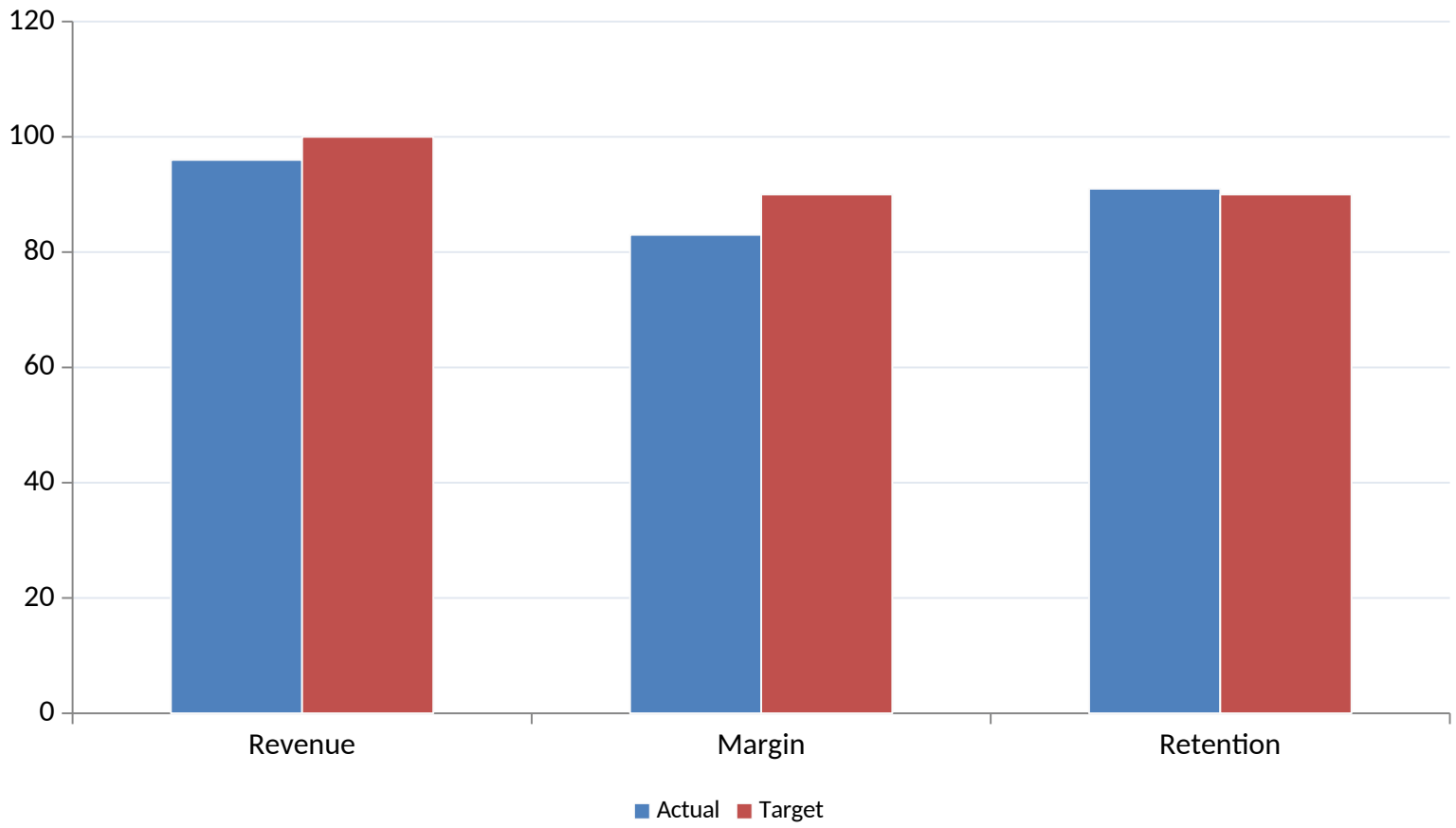
Confirm grain, filter context and whether the display could be misread.



SIGNAL

EDITABLE POWERPOINT CHART

Alerts



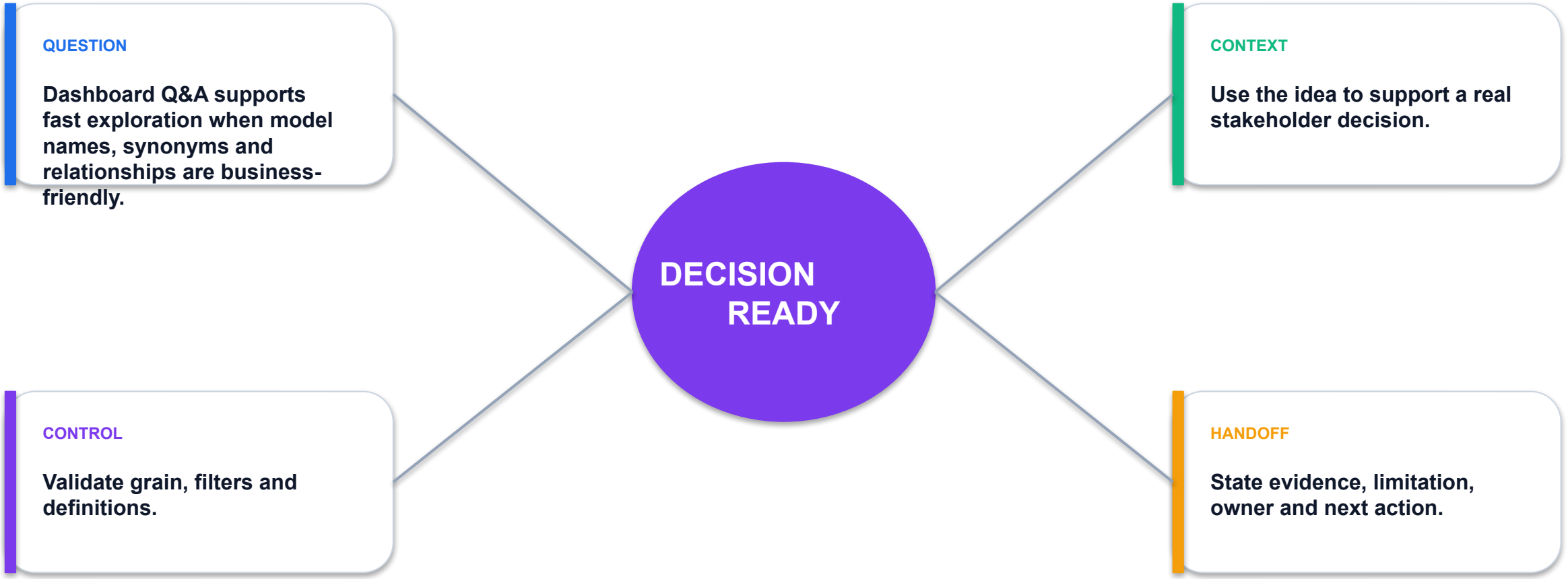
WHAT THE DATA SHOWS

Revenue and margin need an owner and response; retention is above target.

Chart is native and editable in PowerPoint.

CONCEPT MAP

Ask questions



CONCEPT

Quick insights

1

What

Automated patterns are hypotheses to review, not conclusions to accept without context.

2

Why

Use the idea to support a real stakeholder decision.

3

Evidence

Use a defined metric, source and filter state.

4

Action

Translate the finding into an owner and next step.

Dashboard themes

1

Frame

Consistent themes support recognition across tiles without overwhelming status colors.

2

Inspect

Use the idea to support a real stakeholder decision.

3

Test

Check the result under a second filter or scenario.

4

Communicate

State finding, evidence, limitation and action.

CONCEPT

Mobile layout

1

Meaning

Prioritize the few signals and actions needed on a small screen.



SIGNAL

2

Decision use

do not merely shrink desktop design.

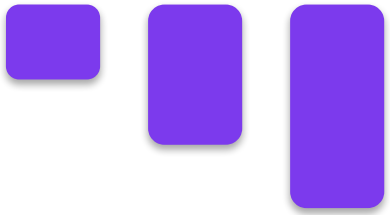


SIGNAL

3

Quality check

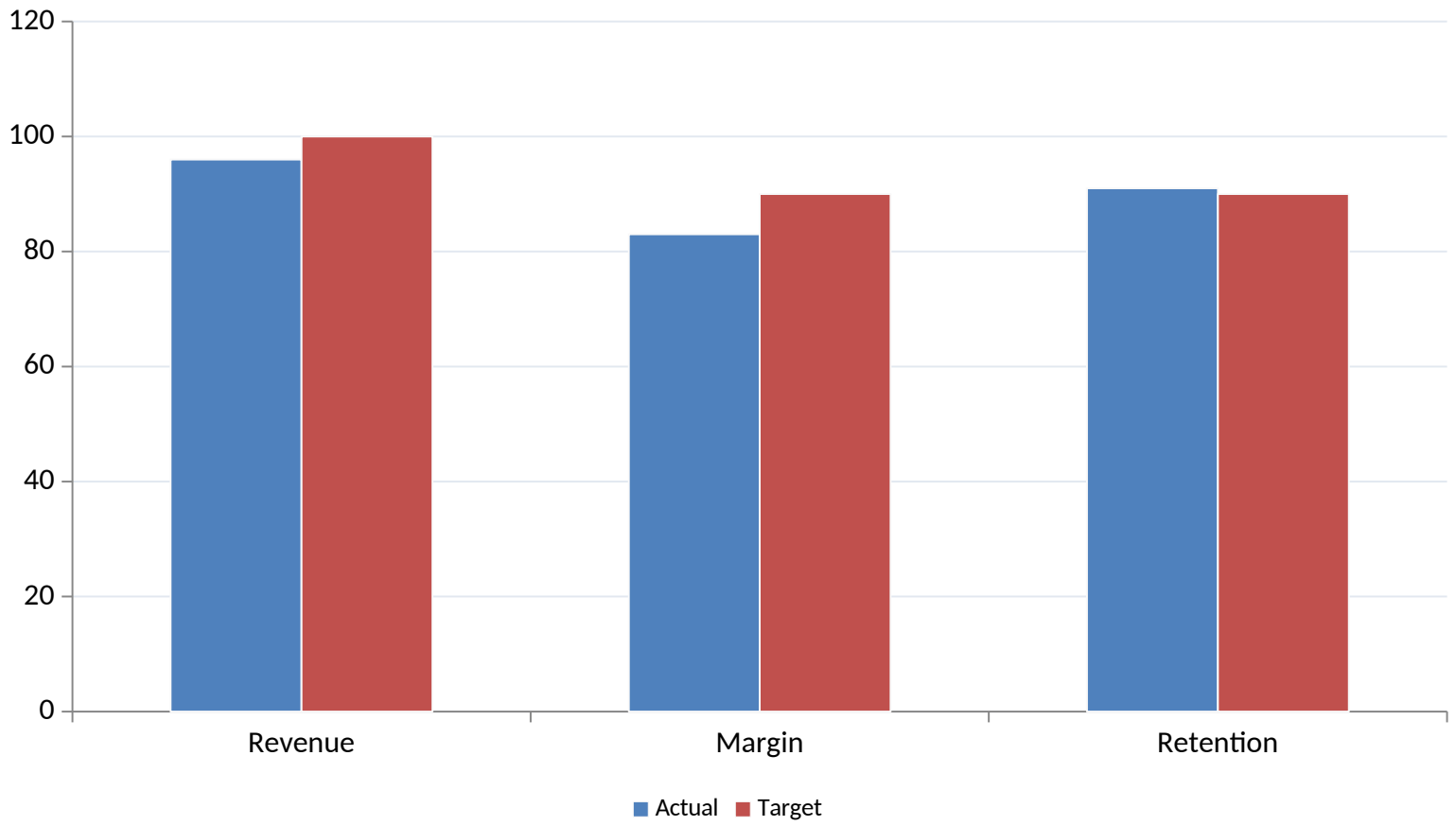
Confirm grain, filter context and whether the display could be misread.



SIGNAL

EDITABLE POWERPOINT CHART

Scorecards and goals



WHAT THE DATA SHOWS

Revenue and margin need an owner and response; retention is above target.

Chart is native and editable in PowerPoint.

CONCEPT

Workspaces

1

What

Workspaces separate authoring, review and distribution responsibilities.

2

Why

Use the idea to support a real stakeholder decision.

3

Evidence

Use a defined metric, source and filter state.

4

Action

Translate the finding into an owner and next step.

CONCEPT

Apps and sharing

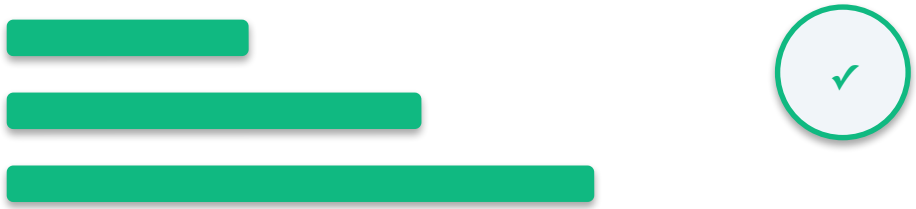
Core idea

Apps package approved content for audiences.



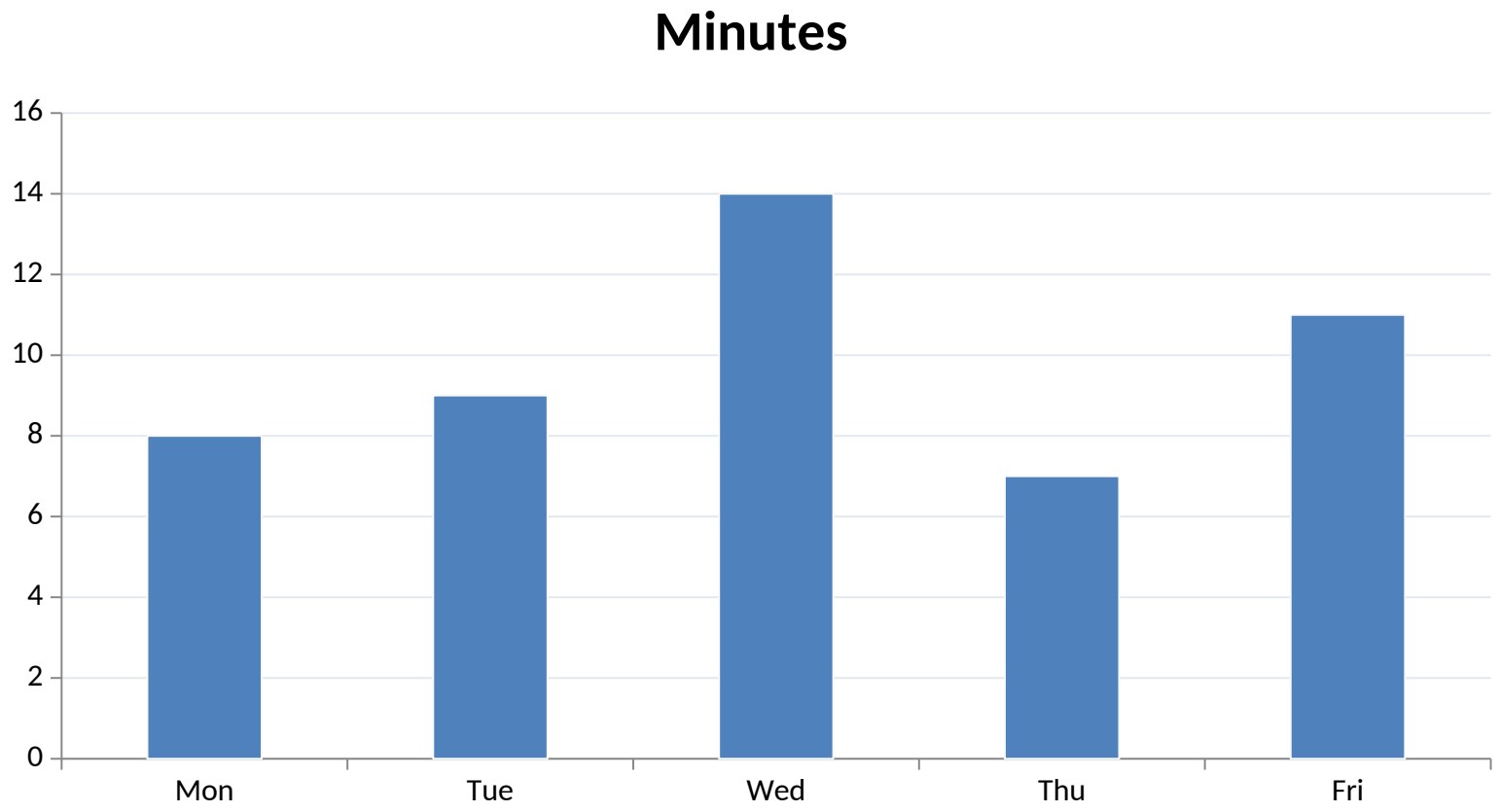
Decision lens

direct sharing is narrower and harder to govern at scale.



EDITABLE POWERPOINT CHART

Refresh and gateways



WHAT THE DATA SHOWS

Wednesday exceeded the refresh service level; investigate source availability and gateway health.

Chart is native and editable in PowerPoint.

CONCEPT

Dashboard storytelling

1

Meaning

Arrange tiles from outcome to drivers to exceptions so executives can scan, diagnose and act.



SIGNAL

2

Decision use

Use the idea to support a real stakeholder decision.

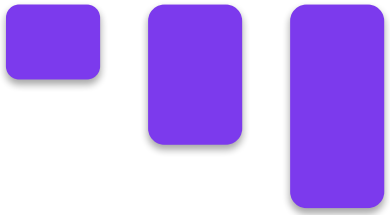


SIGNAL

3

Quality check

Confirm grain, filter context and whether the display could be misread.



SIGNAL

CONCEPT

Dashboard QA

1

What

Verify current data, tile links, mobile view, ownership, permissions and decision usefulness before release.

2

Why

Use the idea to support a real stakeholder decision.

3

Evidence

Use a defined metric, source and filter state.

4

Action

Translate the finding into an owner and next step.

Lab 5 · Build an Executive Dashboard and Scorecard

INTERMEDIATE

05

LO3 · K3 · A3

YOU'LL PRODUCE

A Sales Monitoring dashboard and KPI scorecard in Power BI Service

FILES

Executive_KPIs.xlsx · starter.pbix · README.md/PDF

PASS WHEN

Tiles and targets are current, decision-focused and traceable to the report.

PROCEDURE Learner Guide + lab README

End of Day 1

Save your working PBIX files and complete the Excel evidence sheets.

TOPIC

04

Create Dynamic and Interactive Visualizations

K4 · A4 · LO4

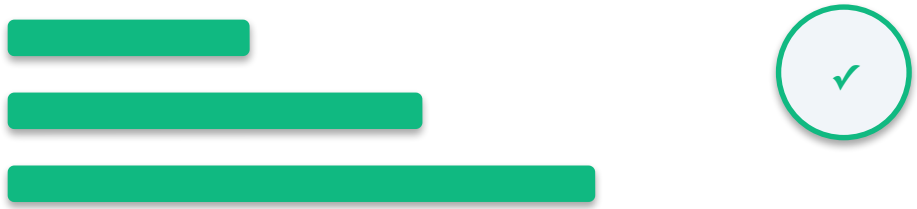
Concepts on slides · detailed procedures in the Learner Guide and lab folders

CONCEPT

Interaction model

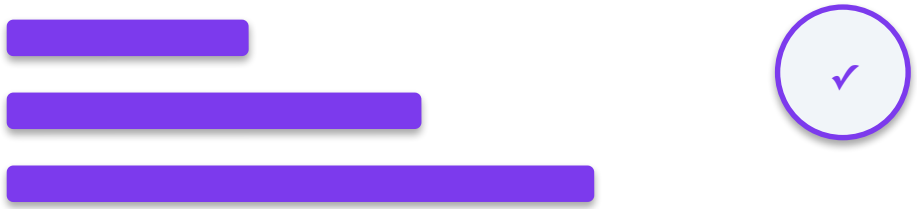
Core idea

Selections propagate through relationships and visual interactions.



Decision lens

every response should be intentional.



Filter scope · Cross-filter

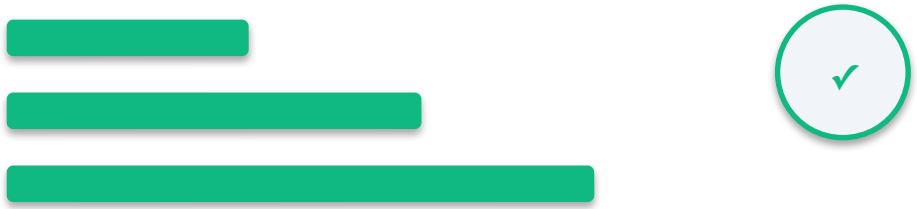
Filter scope

Dataset security, report filters, page filters, visual filters and drillthrough each control a different boundary.



Cross-filter

A selection removes unrelated marks so the page focuses on the chosen segment.



CONCEPT

Cross-highlight

1

What

A selection highlights contribution while retaining the whole for comparison.

2

Why

Use the idea to support a real stakeholder decision.

3

Evidence

Use a defined metric, source and filter state.

4

Action

Translate the finding into an owner and next step.

CONCEPT

Edit interactions

1

Meaning

Configure each visual to filter, highlight or ignore the source selection based on the stakeholder question.



SIGNAL

2

Decision use

Use the idea to support a real stakeholder decision.

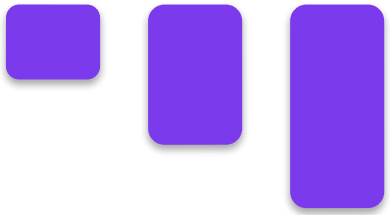


SIGNAL

3

Quality check

Confirm grain, filter context and whether the display could be misread.



SIGNAL

WHY IT MATTERS

Slicers

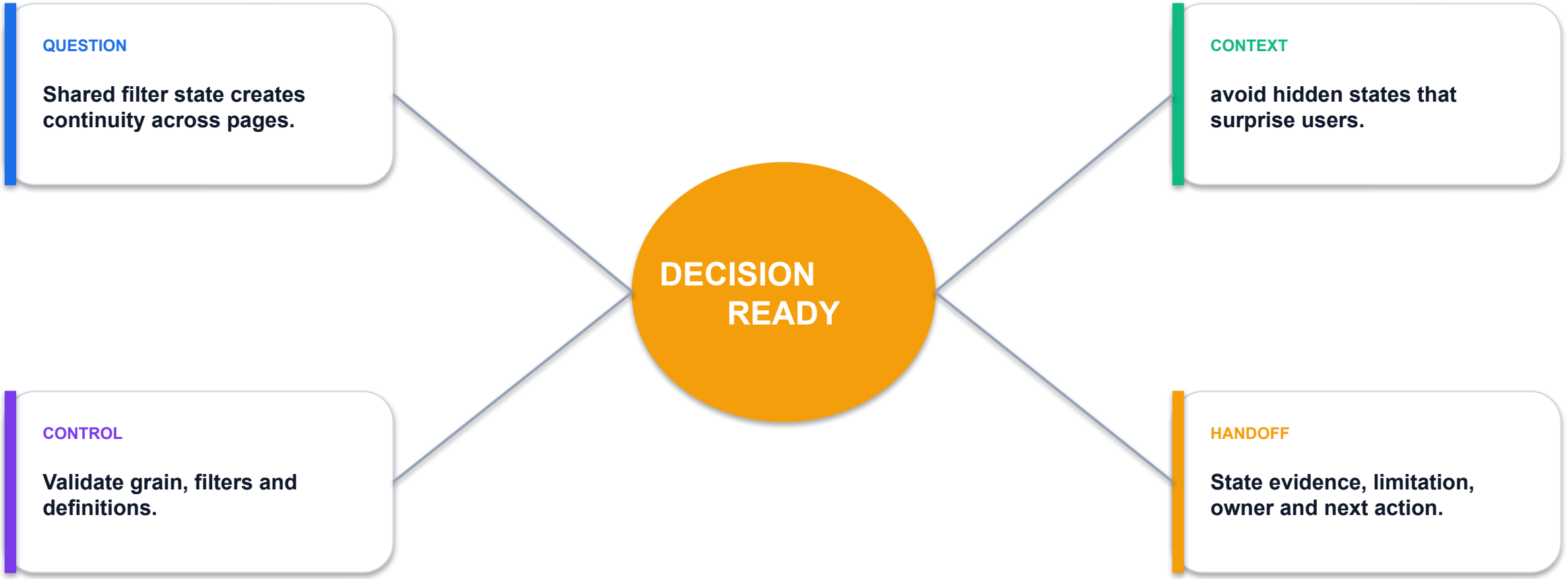
> **Expose the few high-value filters users need and show the active selection clearly.**

Use the idea to support a real stakeholder decision.

QUESTION	What must be decided?
EVIDENCE	What supports the claim?
ACTION	Who acts next?

CONCEPT MAP

Sync slicers



CONCEPT

Hierarchies

1

What

Ordered levels support drill paths such as Year → Quarter → Month or Region → Country → Store.

2

Why

Use the idea to support a real stakeholder decision.

3

Evidence

Use a defined metric, source and filter state.

4

Action

Translate the finding into an owner and next step.

DECISION STEPS

Drillthrough

1. **Frame** — Carry context from a summary mark to a focused detail page with a clear route back.
2. **Inspect** — Use the idea to support a real stakeholder decision.
3. **Test** — Check the result under a second filter or scenario.
4. **Communicate** — State finding, evidence, limitation and action.

CONCEPT

Bookmarks

1

Meaning

Capture filter, display and navigation states to build guided stories and reset controls.



SIGNAL

2

Decision use

Use the idea to support a real stakeholder decision.

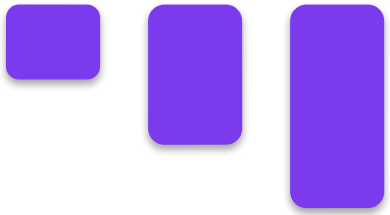


SIGNAL

3

Quality check

Confirm grain, filter context and whether the display could be misread.



SIGNAL

Buttons and navigation

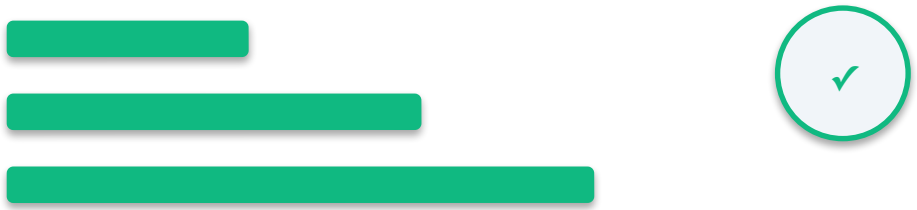
Core idea

Use consistent labels, states and placement so navigation feels predictable.



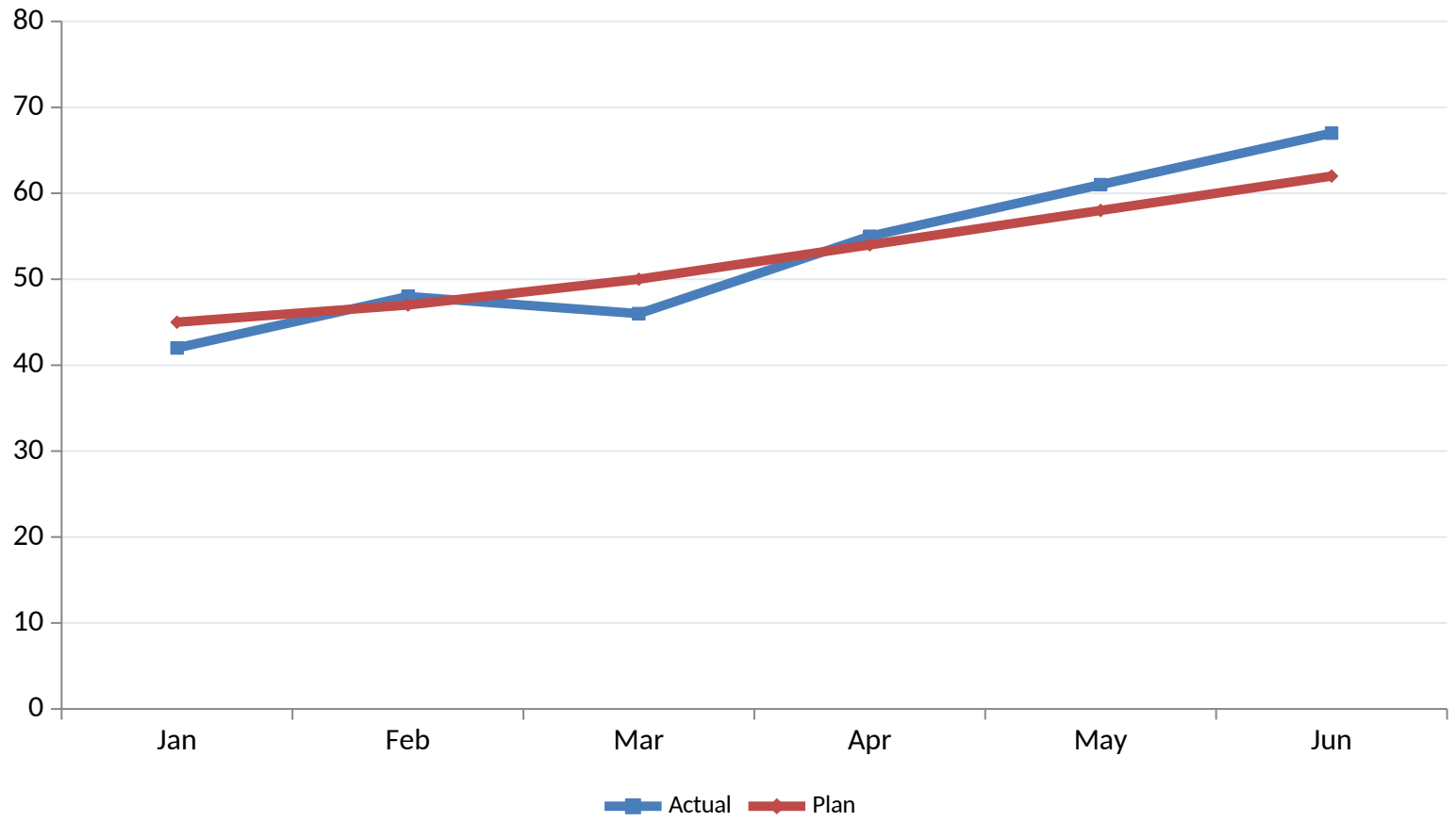
Decision lens

Use the idea to support a real stakeholder decision.



EDITABLE POWERPOINT CHART

Conditional formatting



WHAT THE DATA SHOWS

The editable visual links a measured pattern to a specific review question.

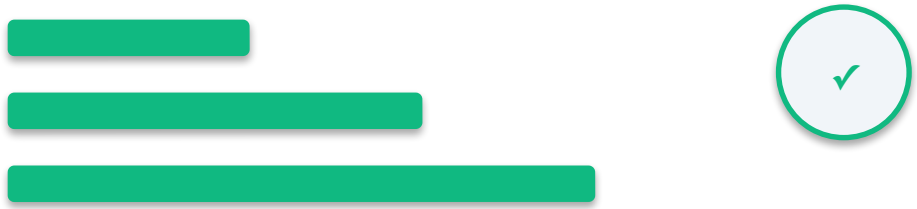
Chart is native and editable in PowerPoint.

CONCEPT

Report-page tooltips

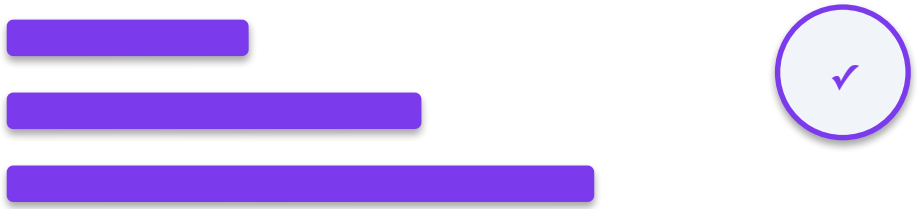
Core idea

Provide rich context on hover while keeping the primary view simple.



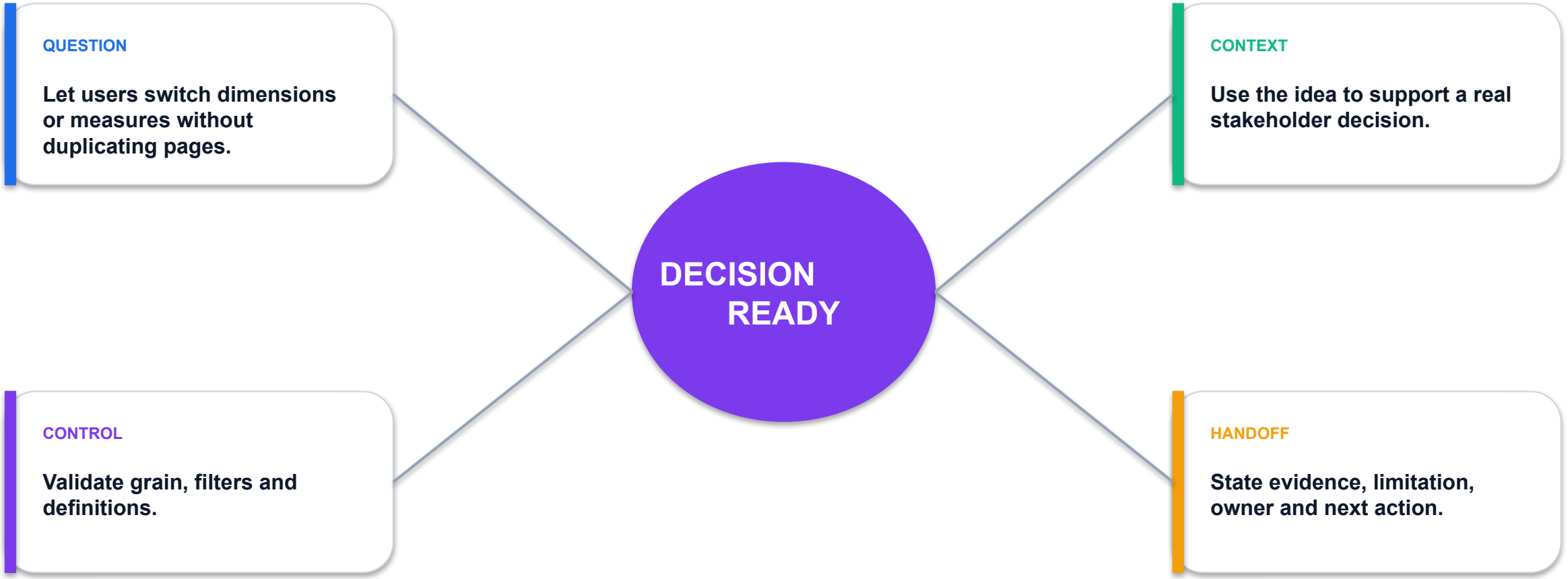
Decision lens

Use the idea to support a real stakeholder decision.



CONCEPT MAP

Field parameters



CONCEPT

What-if parameters

1

Meaning

Turn a report into a controlled scenario tool while distinguishing assumptions from actuals.



SIGNAL

2

Decision use

Use the idea to support a real stakeholder decision.

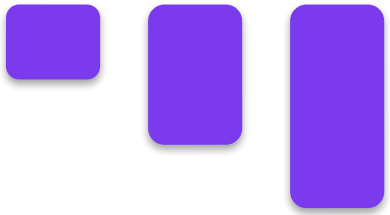


SIGNAL

3

Quality check

Confirm grain, filter context and whether the display could be misread.



SIGNAL

CONCEPT

Interactive UX

1

What

Preserve orientation with page titles, breadcrumbs, reset state and visible filter context.

2

Why

Use the idea to support a real stakeholder decision.

3

Evidence

Use a defined metric, source and filter state.

4

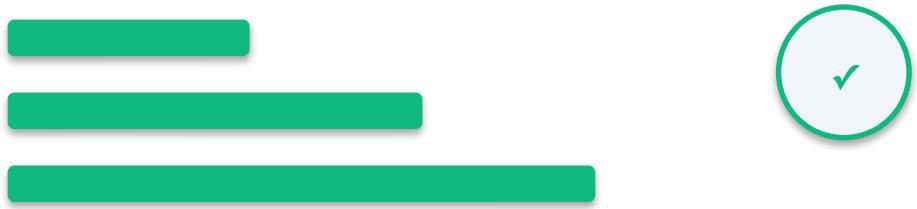
Action

Translate the finding into an owner and next step.

Stakeholder review

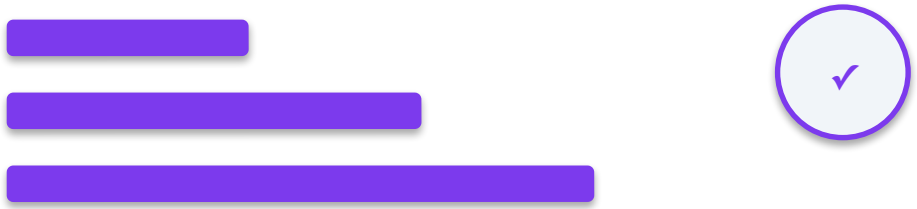
Core idea

Test the display against real questions, expected actions and common misinterpretations.



Decision lens

Use the idea to support a real stakeholder decision.



Lab 6 · Create Dynamic Interactions and Navigation

INTERMEDIATE

06

LO4 · K4 · A4

YOU'LL PRODUCE

An interactive report with slicers, drillthrough, bookmarks and tested interactions

FILES

Campaign_Performance.xlsx · starter.pbix · README.md/PDF

PASS WHEN

Every interaction is intentional; drill paths work; reset navigation restores the default state.

PROCEDURE Learner Guide + lab README

TOPIC

05

Create a Strategic Data Model

K5 · A5 · LO5

Concepts on slides · detailed procedures in the Learner Guide and lab folders

CONCEPT

Semantic model

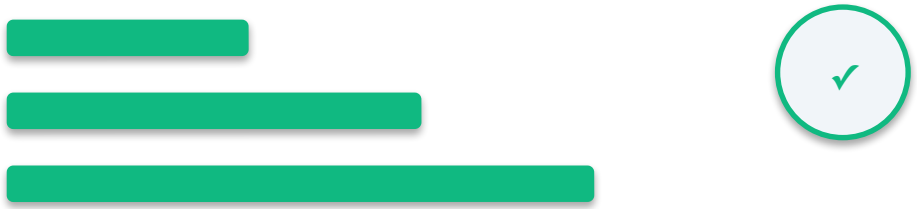
Core idea

A model organizes reusable business meaning: tables, relationships, measures, hierarchies and security.



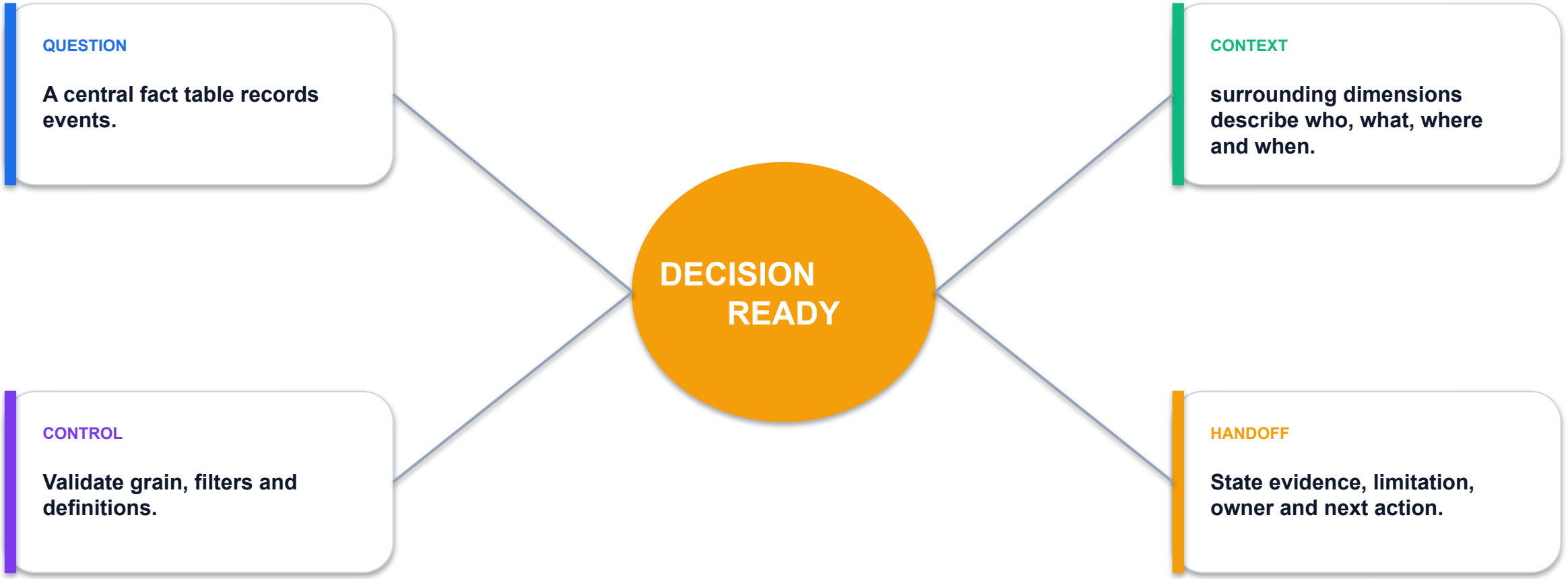
Decision lens

Use the idea to support a real stakeholder decision.



CONCEPT MAP

Star schema



WHY IT MATTERS

Fact tables

> **Facts have a declared grain, foreign keys and numeric observations suitable for aggregation.**

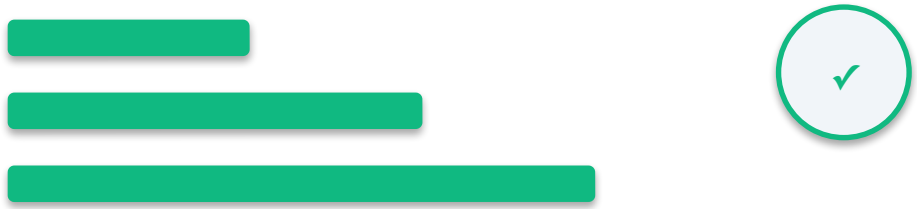
Use the idea to support a real stakeholder decision.

QUESTION	What must be decided?
EVIDENCE	What supports the claim?
ACTION	Who acts next?

Dimension tables · Relationship keys

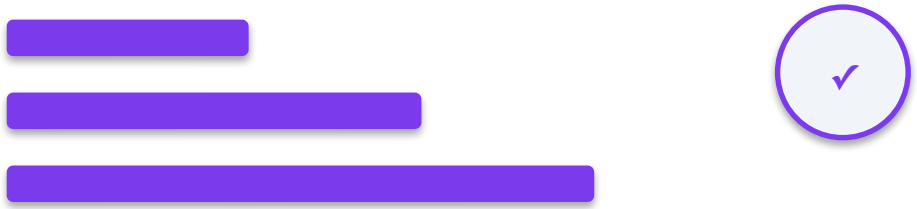
Dimension tables

Dimensions provide stable descriptive attributes for grouping, filtering and labeling.



Relationship keys

Use unique dimension keys and matching fact foreign keys; validate unmatched and duplicate values.



WHY IT MATTERS

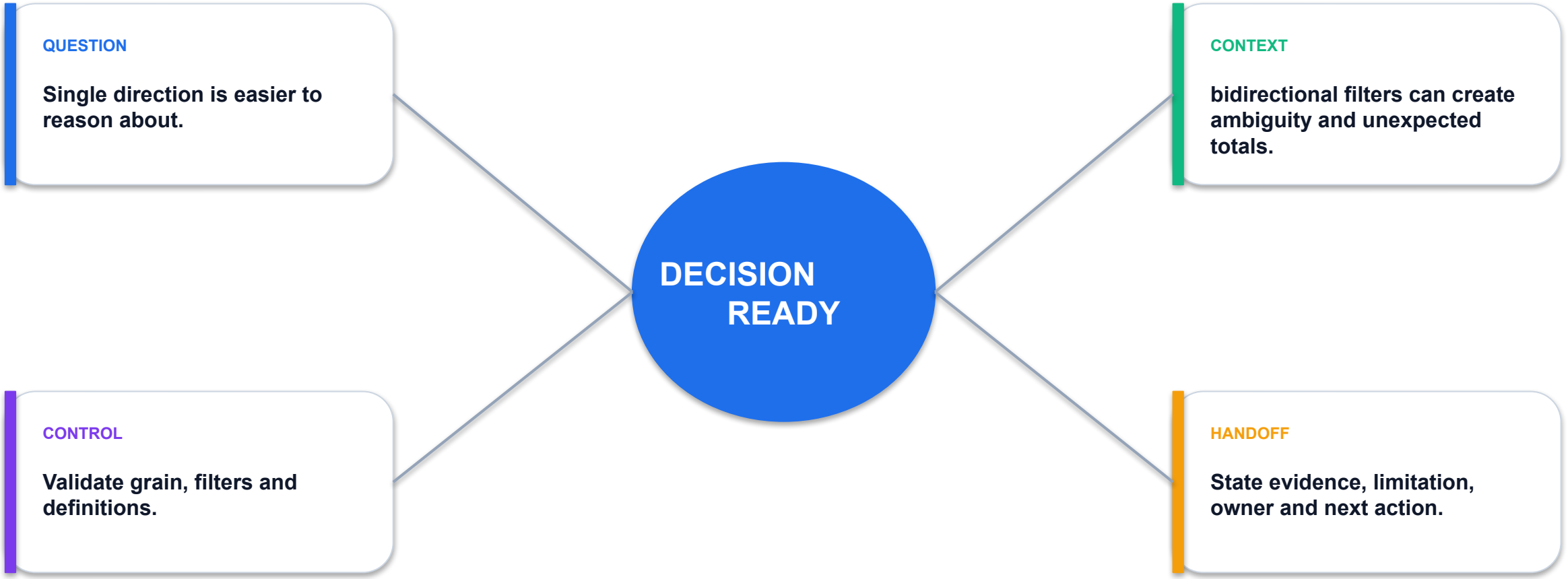
Cardinality

> **One-to-many is the normal star pattern.**
>
> one-to-one and many-to-many require explicit justification.

QUESTION	What must be decided?
EVIDENCE	What supports the claim?
ACTION	Who acts next?

CONCEPT MAP

Cross-filter direction



CONCEPT

Many-to-many

1

What

Bridge tables make shared membership explicit and prevent uncontrolled duplication.

2

Why

Use the idea to support a real stakeholder decision.

3

Evidence

Use a defined metric, source and filter state.

4

Action

Translate the finding into an owner and next step.

DECISION STEPS

Date dimension

1

Frame

A contiguous date table enables consistent calendar attributes and time intelligence.

2

Inspect

Use the idea to support a real stakeholder decision.

3

Test

Check the result under a second filter or scenario.

4

Communicate

State finding, evidence, limitation and action.

CONCEPT

Table and column properties

1

Meaning

Names, formats, data categories and default summarization shape the authoring experience.



SIGNAL

2

Decision use

Use the idea to support a real stakeholder decision.

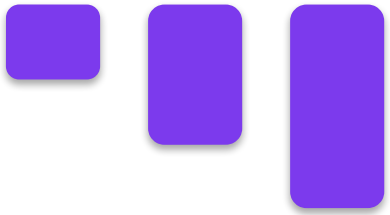


SIGNAL

3

Quality check

Confirm grain, filter context and whether the display could be misread.



SIGNAL

CONCEPT

Hierarchies

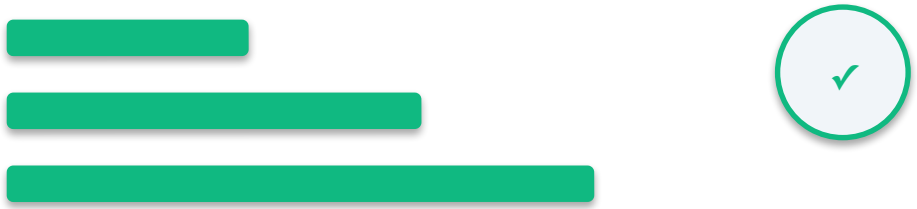
Core idea

Curated hierarchies guide valid drill paths and reduce repeated setup.



Decision lens

Use the idea to support a real stakeholder decision.



CONCEPT

Measures

1

What

Measures calculate at query time and respond to filter context.

2

Why

Use the idea to support a real stakeholder decision.

3

Evidence

Use a defined metric, source and filter state.

4

Action

Translate the finding into an owner and next step.

CONCEPT

Calculated columns

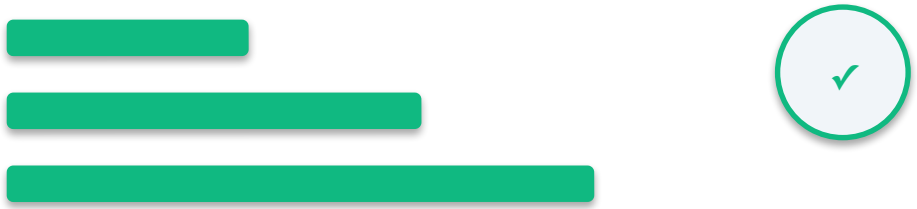
Core idea

Columns calculate row by row at refresh and increase model size.



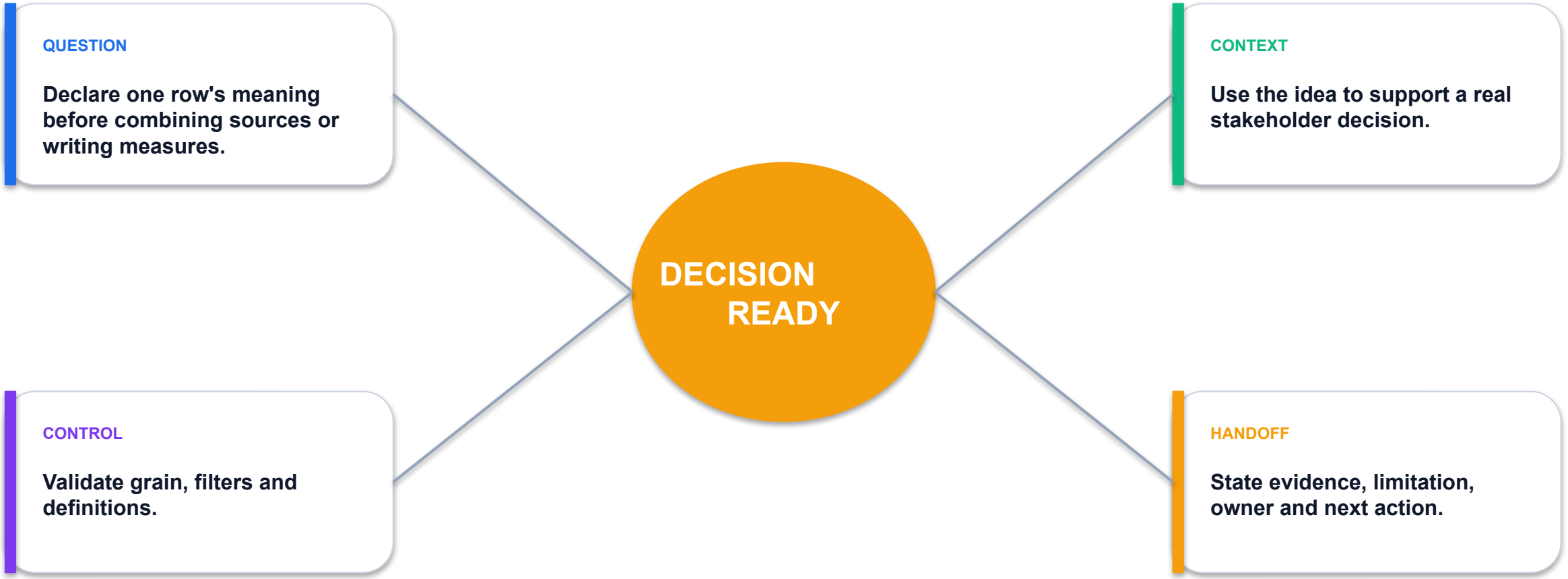
Decision lens

use them only when a stored attribute is needed.



CONCEPT MAP

Granularity



CONCEPT

Role-playing dimensions

1

Meaning

One Date table can relate to order, ship and due dates.



SIGNAL

2

Decision use

only one relationship is active at a time.

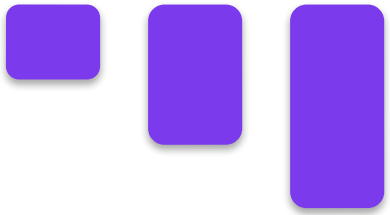


SIGNAL

3

Quality check

Confirm grain, filter context and whether the display could be misread.



SIGNAL

CONCEPT

Model performance

1

What

Remove unused columns, prefer star schemas, reduce high cardinality and keep transformations upstream where practical.

2

Why

Use the idea to support a real stakeholder decision.

3

Evidence

Use a defined metric, source and filter state.

4

Action

Translate the finding into an owner and next step.

Business-friendly model

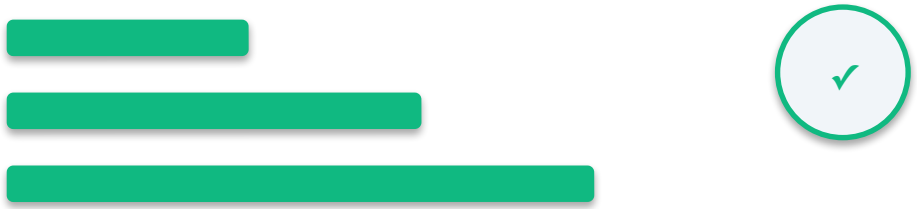
Core idea

Hide technical keys, group measures and use descriptions so authors find the right fields.



Decision lens

Use the idea to support a real stakeholder decision.



Model QA

1. **Frame** — Check uniqueness, referential integrity, cardinality, filter paths, totals and refresh behavior before visualization.
2. **Inspect** — Use the idea to support a real stakeholder decision.
3. **Test** — Check the result under a second filter or scenario.
4. **Communicate** — State finding, evidence, limitation and action.

Lab 7 · Model a Star Schema and Date Dimension

ADVANCED

07

LO5 · K5 · A5

YOU'LL PRODUCE

A semantic model with fact/dimension relationships and a marked date table

FILES

Star_Schema.xlsx · starter.pbix · README.md/PDF

PASS WHEN

Relationships use correct keys/cardinality; filters flow predictably; model ambiguity is zero.

PROCEDURE Learner Guide + lab README

TOPIC

06

Analyze Data with DAX

A6 · LO6

Concepts on slides · detailed procedures in the Learner Guide and lab folders

DAX purpose

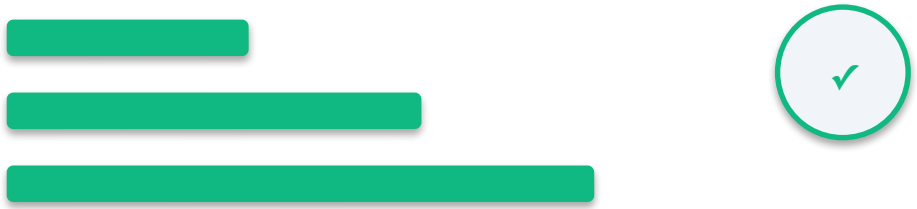
Core idea

DAX defines reusable calculations over a semantic model.



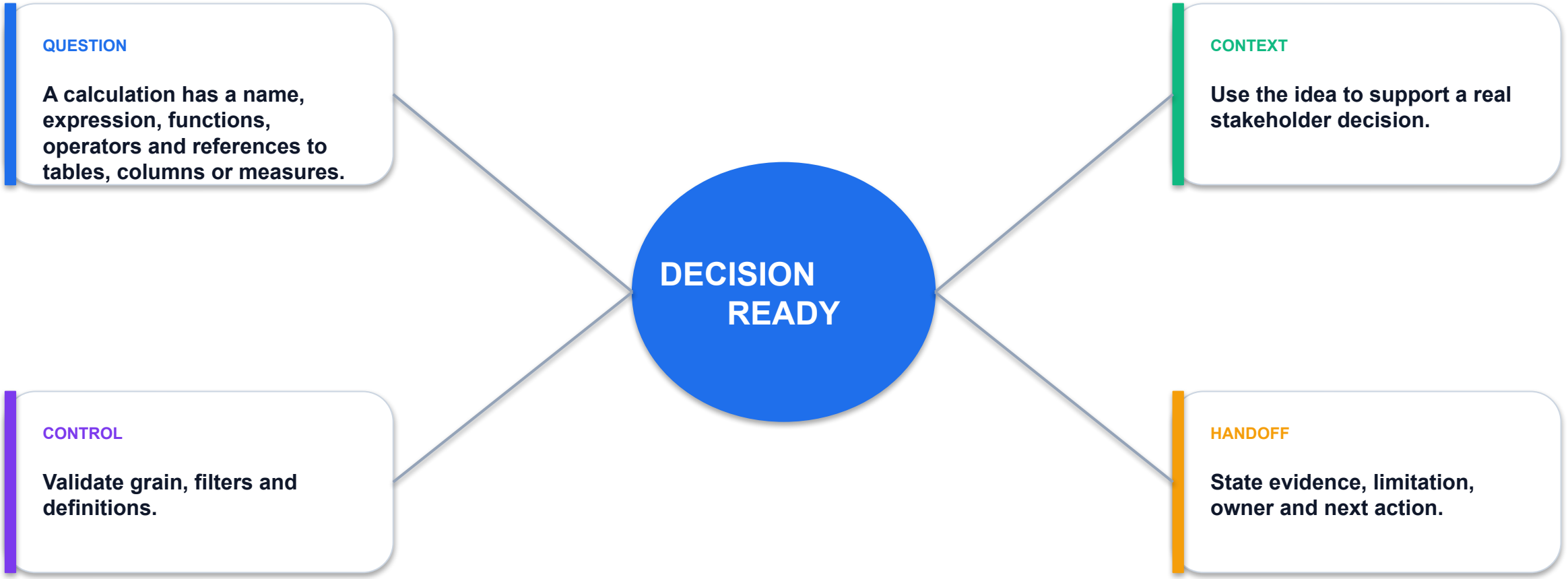
Decision lens

it is not a replacement for data cleaning.



CONCEPT MAP

Formula syntax



WHY IT MATTERS

Reference rules

> **Columns require table context.**
>
> measures should be referenced by measure name to preserve reusable logic.

QUESTION	What must be decided?
EVIDENCE	What supports the claim?
ACTION	Who acts next?

CONCEPT

Data types and operators

1

What

Type compatibility controls comparison, arithmetic and blank handling.

2

Why

Use the idea to support a real stakeholder decision.

3

Evidence

Use a defined metric, source and filter state.

4

Action

Translate the finding into an owner and next step.

Row context · Filter context

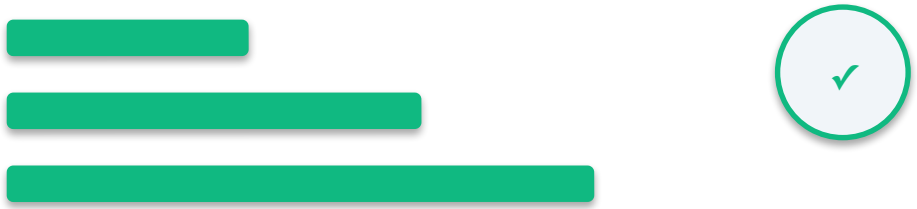
Row context

Calculated columns and iterators evaluate one row at a time.



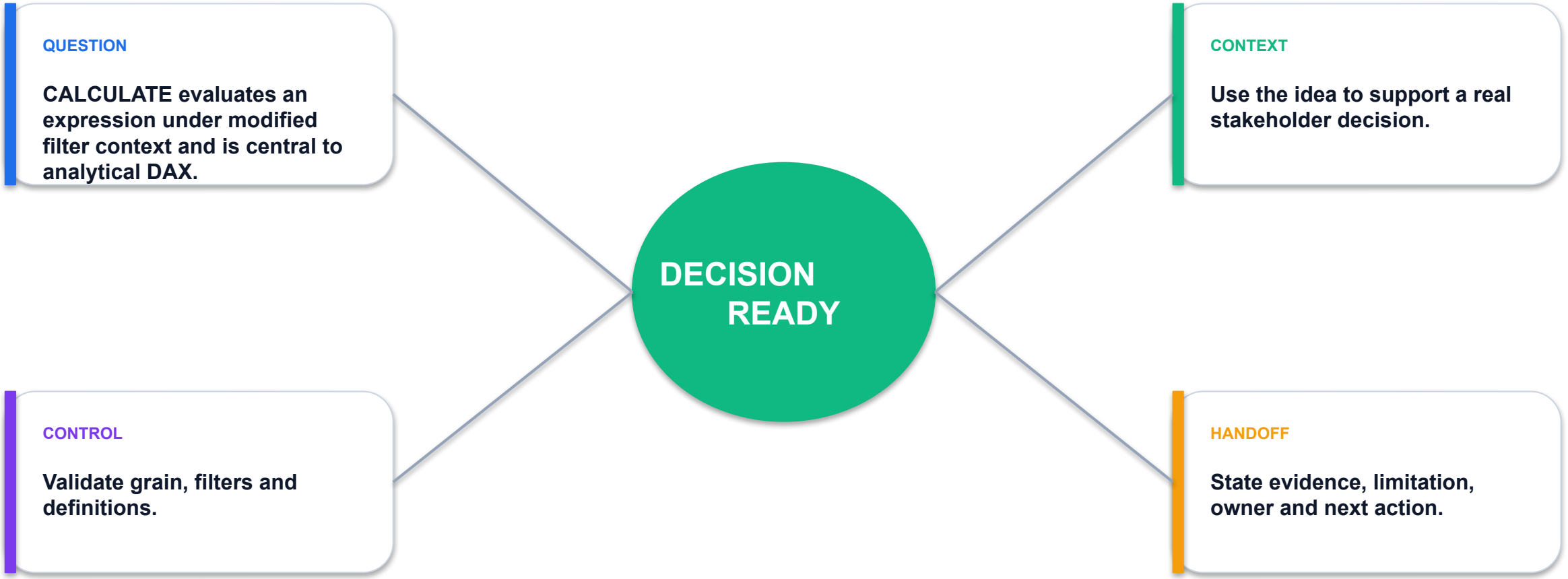
Filter context

Visual axes, slicers, filters and CALCULATE define the subset a measure evaluates.



CONCEPT MAP

CALCULATE



CONCEPT

Variables

1

What

VAR and RETURN improve readability, avoid repeated evaluation and simplify debugging.

2

Why

Use the idea to support a real stakeholder decision.

3

Evidence

Use a defined metric, source and filter state.

4

Action

Translate the finding into an owner and next step.

Implicit and explicit measures

1. **Frame** — Explicit measures centralize definitions, formatting and governance.
2. **Inspect** — Use the idea to support a real stakeholder decision.
3. **Test** — Check the result under a second filter or scenario.
4. **Communicate** — State finding, evidence, limitation and action.

CONCEPT

Core measures

1

Meaning

Start with additive Sales, Cost and Quantity, then derive Profit and Margin from governed measures.



SIGNAL

2

Decision use

Use the idea to support a real stakeholder decision.

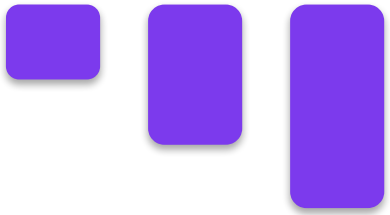


SIGNAL

3

Quality check

Confirm grain, filter context and whether the display could be misread.



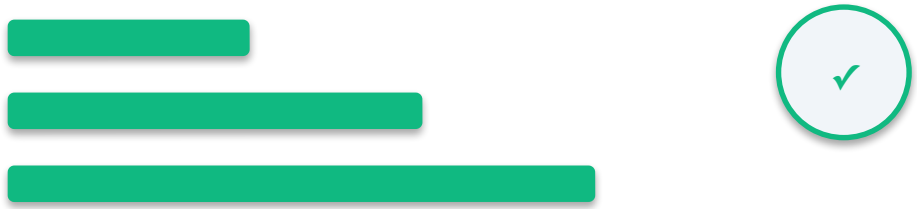
SIGNAL

CONCEPT

Safe division

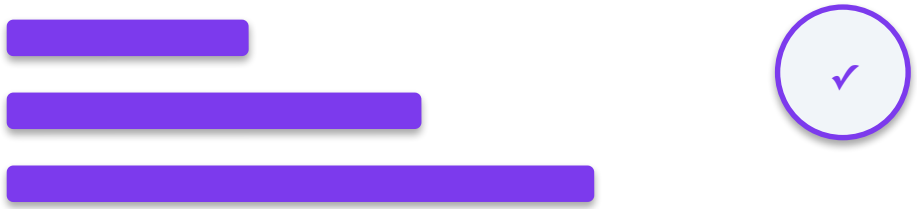
Core idea

DIVIDE handles zero or blank denominators more safely than the division operator.



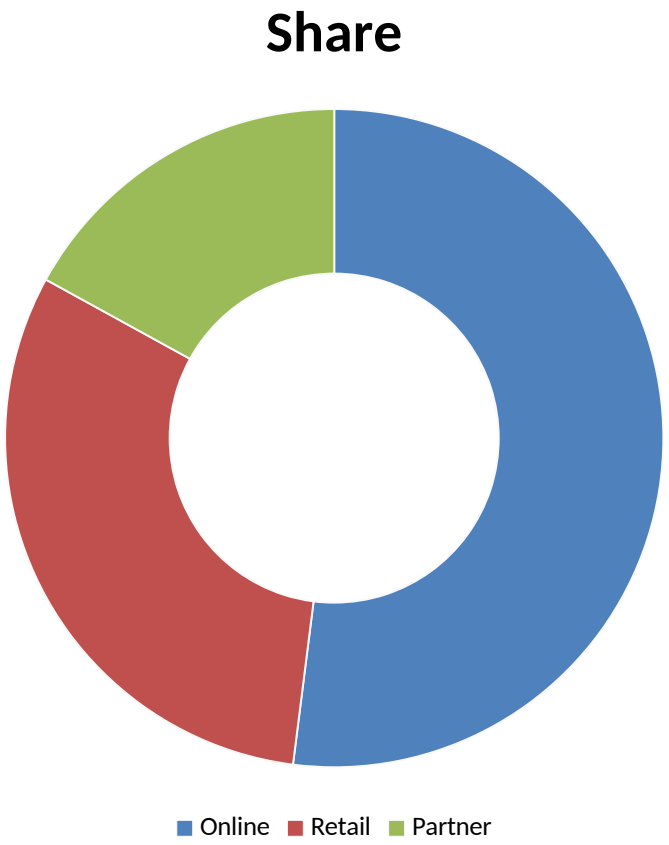
Decision lens

Use the idea to support a real stakeholder decision.



EDITABLE POWERPOINT CHART

Share of total



WHAT THE DATA SHOWS

Use part-to-whole only when the total is meaningful and categories are few.

Chart is native and editable in PowerPoint.

Date table

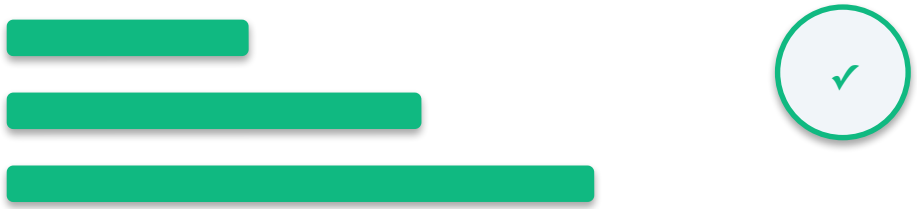
Core idea

A marked date table with continuous dates underpins consistent time calculations.



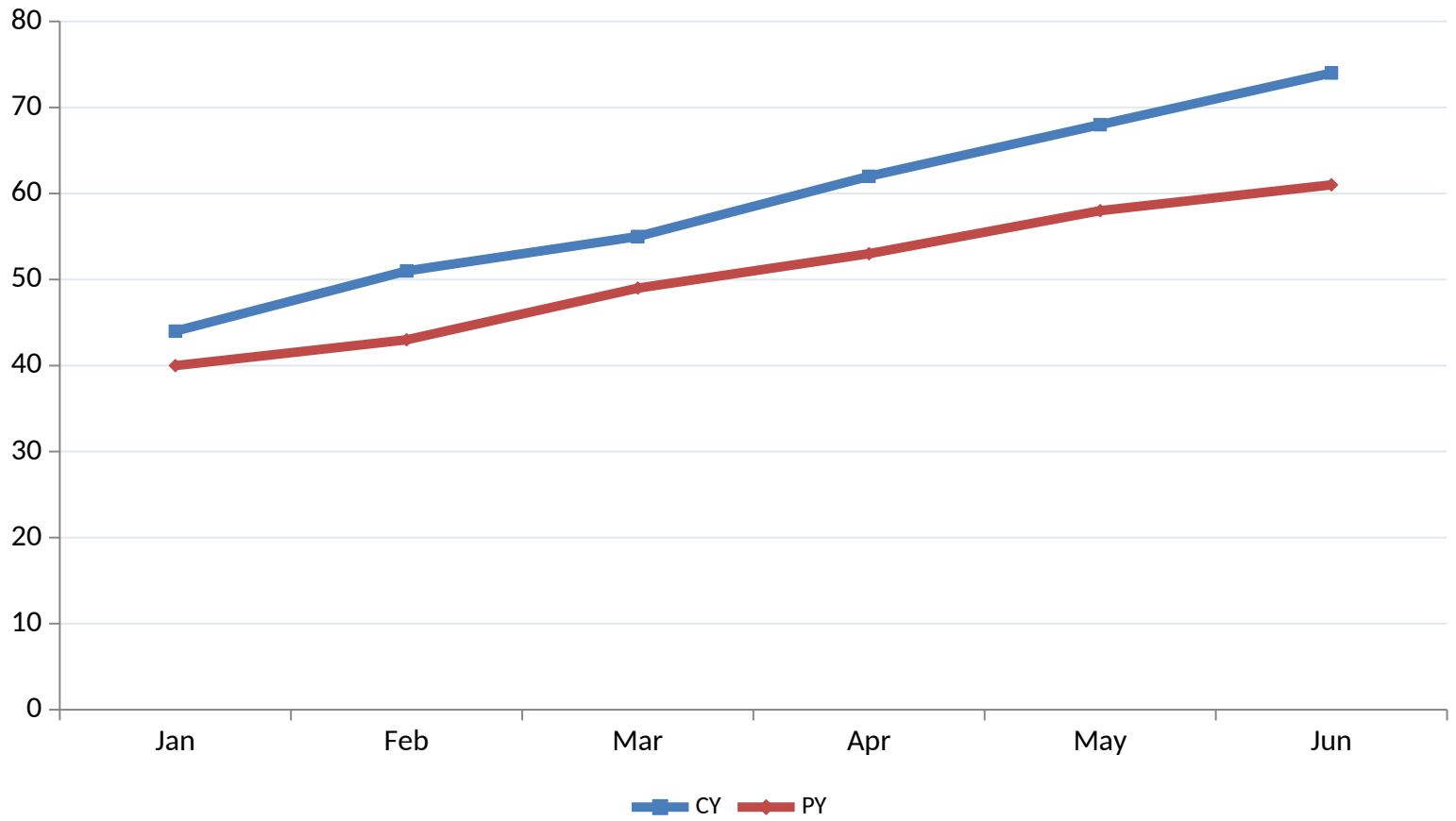
Decision lens

Use the idea to support a real stakeholder decision.



EDITABLE POWERPOINT CHART

Time intelligence



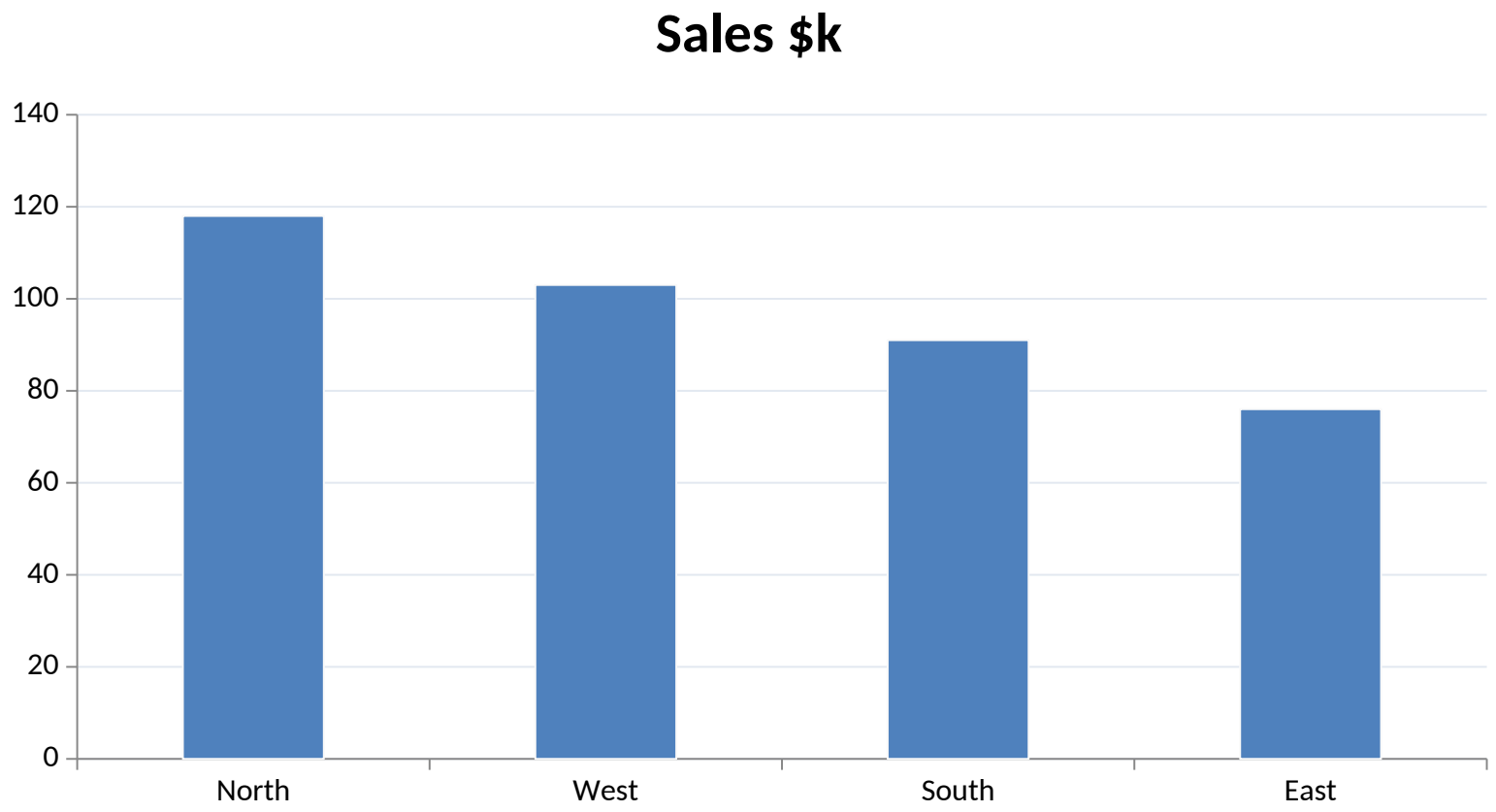
WHAT THE DATA SHOWS

Current-year growth is positive in every month; verify calendar logic before acting on the gap.

Chart is native and editable in PowerPoint.

EDITABLE POWERPOINT CHART

TOPN



WHAT THE DATA SHOWS

A common baseline and descending order make the ranking immediately comparable.

Chart is native and editable in PowerPoint.

CONCEPT

Groups and bins

1

What

Grouping creates analytical segments.

2

Why

bins reveal distribution at a chosen resolution.

3

Evidence

Use a defined metric, source and filter state.

4

Action

Translate the finding into an owner and next step.

Outliers and anomalies

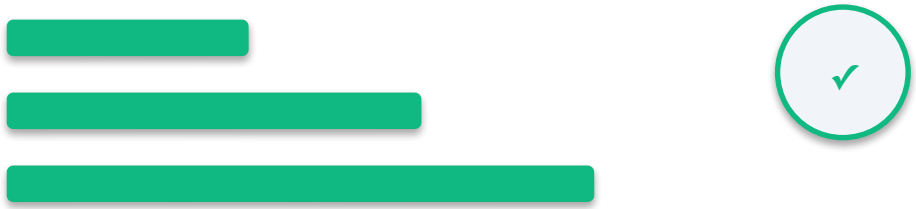
Core idea

Flag exceptions using statistical context, then investigate causes rather than treating detection as explanation.



Decision lens

Use the idea to support a real stakeholder decision.



Clustering

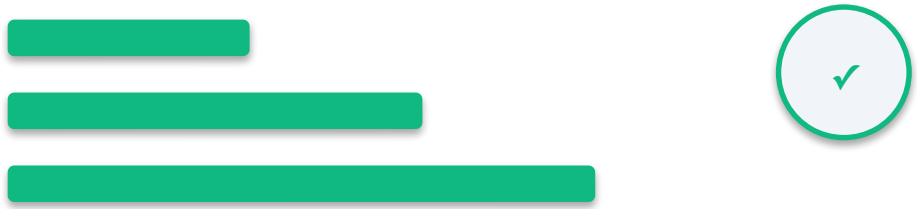
1. **Frame** — Clusters suggest similar records based on selected variables.
2. **Inspect** — business review is needed to name and use them.
3. **Test** — Check the result under a second filter or scenario.
4. **Communicate** — State finding, evidence, limitation and action.

CONCEPT

Analyze feature

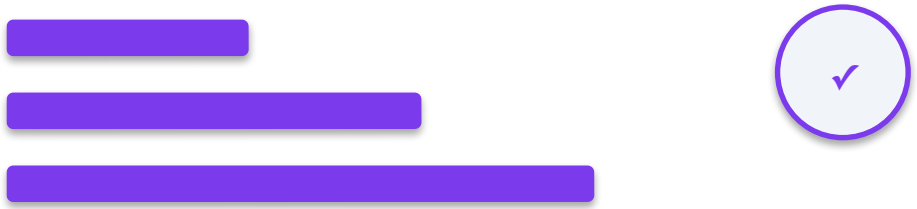
Core idea

Automated increase/decrease explanations provide candidate drivers that require validation.



Decision lens

Use the idea to support a real stakeholder decision.



CONCEPT

What-if analysis

1

Meaning

Disconnected parameter tables let users test assumptions without changing actual data.



SIGNAL

2

Decision use

Use the idea to support a real stakeholder decision.

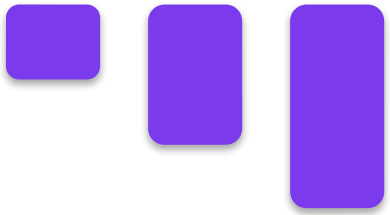


SIGNAL

3

Quality check

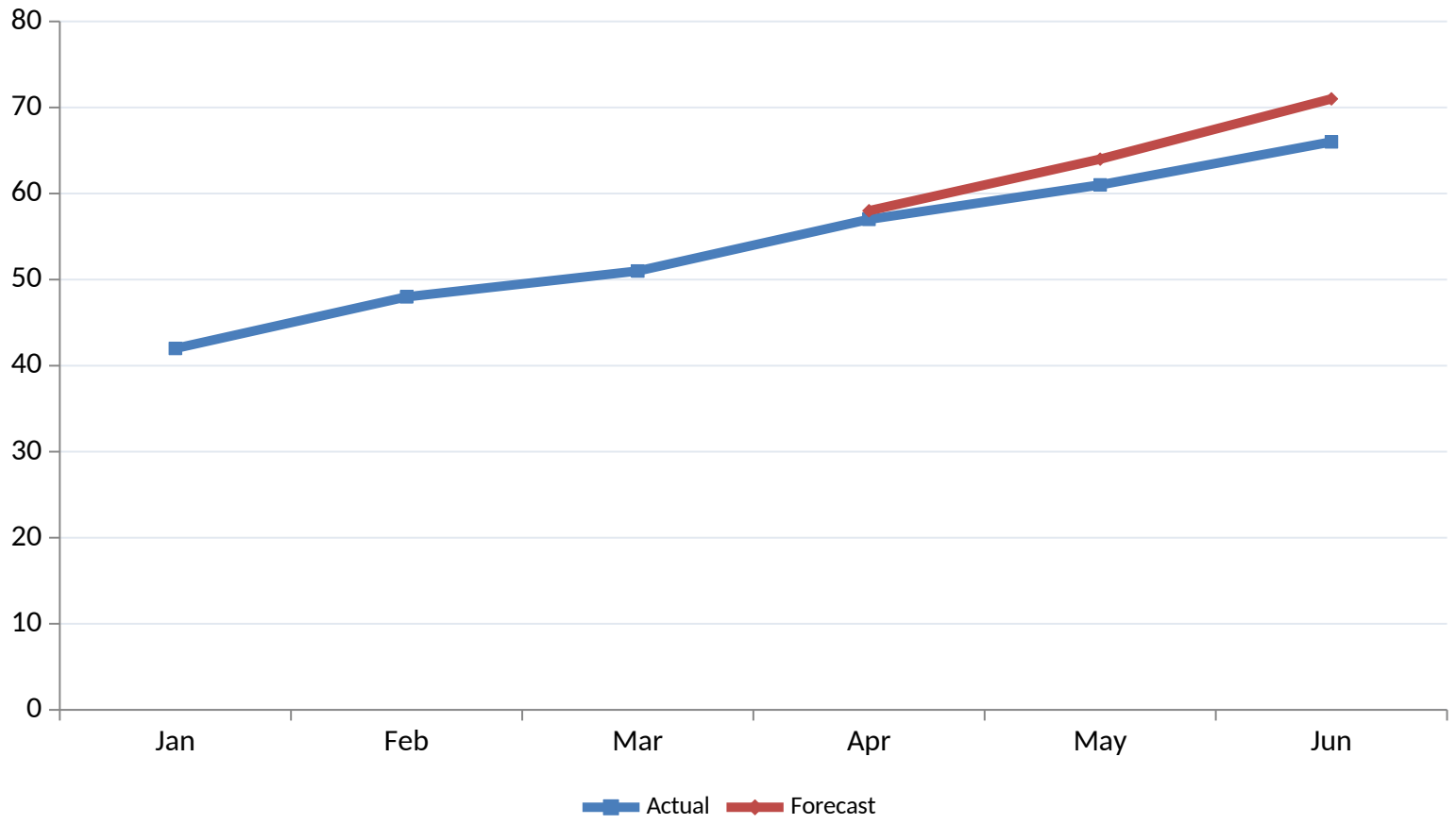
Confirm grain, filter context and whether the display could be misread.



SIGNAL

EDITABLE POWERPOINT CHART

Trend lines and forecasts



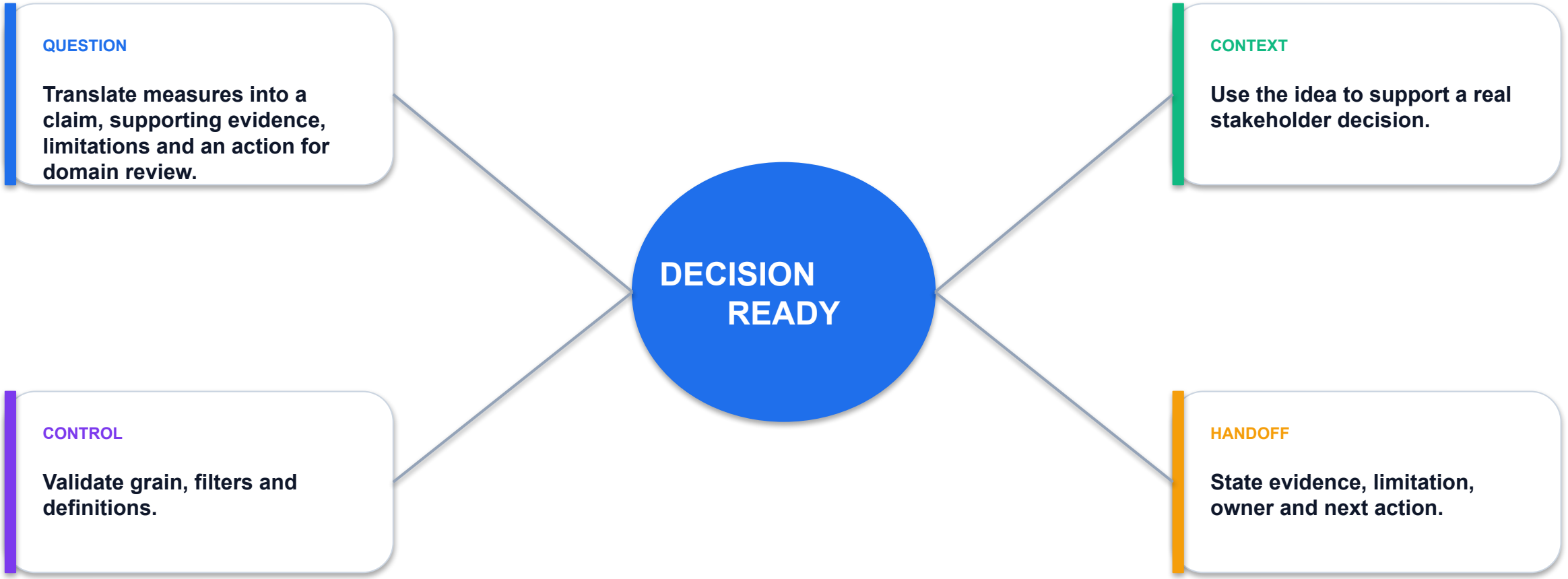
WHAT THE DATA SHOWS

The observed series rises; the forecast extends direction with uncertainty that still needs business context.

Chart is native and editable in PowerPoint.

CONCEPT MAP

Align findings with SMEs



Lab 8 · Create Core DAX Measures

ADVANCED

08

LO6 · A6

YOU'LL PRODUCE

Validated measures for Sales, Cost, Profit, Margin and share of total

FILES

Sales_Model.xlsx · starter.pbix · README.md/PDF

PASS WHEN

Measures reconcile to workbook controls and respond correctly to filter context.

PROCEDURE Learner Guide + lab README

Lab 9 · Apply Time Intelligence and Scenario Analysis

ADVANCED

09

LO6 · A6

YOU'LL PRODUCE

YTD, prior-period, Top N and what-if analysis with decision commentary

FILES

Scenario_Analysis.xlsx · starter.pbix · README.md/PDF

PASS WHEN

Date logic is valid; parameters change results; findings state business meaning and limitations.

PROCEDURE Learner Guide + lab README

Lab 10 · Deliver the GlobalTech Sales Performance Cockpit

CAPSTONE

10

LO1–LO6 · K1–K5 · A1–A6

YOU'LL PRODUCE

A stakeholder-ready PBIX report, dashboard evidence and QA checklist

FILES

GlobalTech_Capstone.xlsx · starter.pbix · README.md/PDF

PASS WHEN

The solution meets all six outcomes, answers stakeholder questions and passes the evidence checklist.

PROCEDURE Learner Guide + lab README

Capstone · From Evidence to Action

DECISION MOMENT

GlobalTech leaders need one defensible performance cockpit: prepare trustworthy data, model it clearly, calculate governed measures and communicate action.

Concept visual · validate every conclusion with the supplied data.



Capstone Acceptance Gate

PASS THE EVIDENCE GATE

- ✓ Refresh succeeds and transformations are documented
- ✓ Star schema and relationships pass integrity checks
- ✓ DAX totals reconcile to Excel controls
- ✓ Every visual answers a stakeholder question
- ✓ Interactions, accessibility and mobile layout are tested

What You Achieved

1

LO1

Perform data transformation techniques using Power BI

2

LO2

Create data visualizations that portray patterns, trends and correlations

3

LO3

Develop dashboards and scorecards for decision monitoring

4

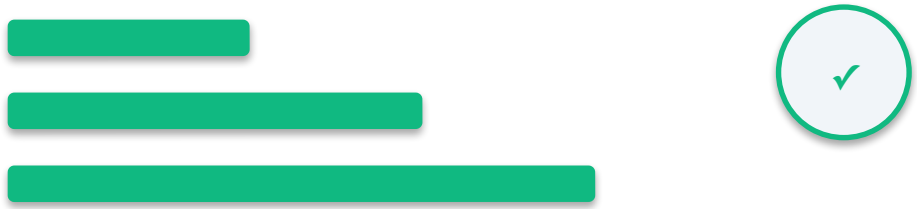
LO4

Create dynamic and interactive data visualizations

What You Achieved · Model and Analyze

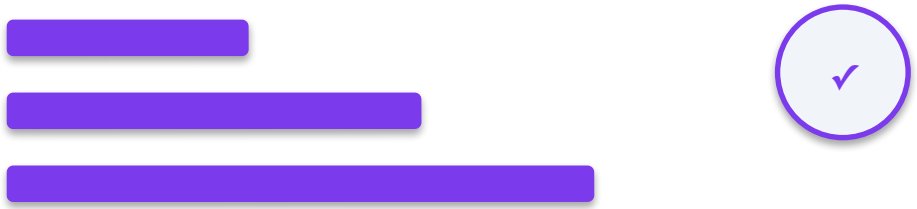
LO5

Create a data model for strategic visualization and mapping



LO6

Perform data analytics and queries with DAX



Courseware and Assessment on the LMS

1

Download

Slides, Learner Guide and lab materials

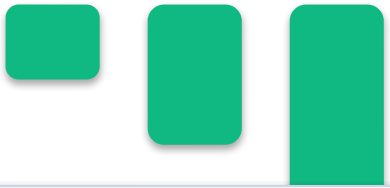


SIGNAL

2

Complete

WA then Practical Performance

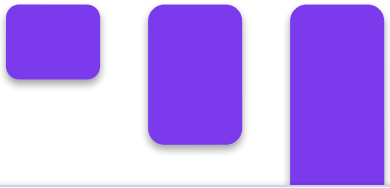


SIGNAL

3

Submit

Upload completed answers and evidence



SIGNAL

https://lms-tms.tertiaryinfotech.com/

FINAL REMINDER

Assessment

1

WA

5 open-ended knowledge questions · 60 min



SIGNAL

2

PP

5 applied tasks · 90 min

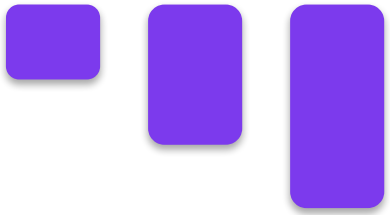


SIGNAL

3

Submit

Assessment answers and evidence on the LMS



SIGNAL

Assessment Flow



Digital Attendance (Mandatory)

1

Assessment

Complete the SSG assessment attendance

2

TRAQOM

Scan the trainer-provided QR code

3

Submit

Confirm the attendance response

4

Record

Then complete the Assessment Summary Record

THANK YOU

Turn evidence into a clear decision — and make the reasoning reviewable.

enquiry@tertiaryinfotech.com · +65 6100 0613 · www.tertiarycourses.com.sg